

# **The Blind Spot Paradox**

## **Formalizing the Race and Measuring the Adaptation**

Alexandre Boyer   Melaine Gouillou  
Salomé Fonvielle   Ulysse Petit-Tichanné

Filière Recherche — Subject 39

Placeholder.

## NOTATION USED THROUGHOUT

|                                       |                                                                                                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $\tau^*$                              | the drift instant, known by construction                                                                                                       |
| $\tau_{\text{ARF}}$                   | delay until the first tree is replaced, $\min_i \tau_i$ over the $M$ trees                                                                     |
| $\tau_{\text{det}}$                   | delay until the first external alarm, possibly $+\infty$                                                                                       |
| $F_A, F_D$                            | cumulative distribution functions of $\tau_{\text{ARF}}$ and $\tau_{\text{det}}$                                                               |
| $H$                                   | post-drift observation horizon, $H = 2000$                                                                                                     |
| $\lambda, \delta_P$                   | the external detector's threshold and tolerance                                                                                                |
| $\Delta e$                            | drift amplitude, the error jump a non-adapting model would suffer                                                                              |
| Miss                                  | the “blind spot” event, defined in Section I-A                                                                                                 |
| <i>for the inter-tree analysis:</i>   |                                                                                                                                                |
| $M = 10$                              | number of trees in the forest                                                                                                                  |
| $\tau_i$                              | adaptation delay of tree $i$ , so $\tau_{\text{ARF}} = \min_i \tau_i$                                                                          |
| $S$                                   | the complete realization of the stream (one seed)                                                                                              |
| $U_i$                                 | the randomness specific to tree $i$ , identified in Section II-A                                                                               |
| $F$                                   | CDF of a <i>single</i> tree $\tau_i$ — distinct from $F_A$ , which is that of $\tau_{\text{ARF}}$                                              |
| $G_S(s)$                              | $:= P(\tau_1 > s \mid S)$ , the single-tree survival function <i>given</i> the stream                                                          |
| $\beta$                               | reliability target, the share of drifts the monitor should catch (0.95 in the manuscript)                                                      |
| <i>for the instrumented campaign:</i> |                                                                                                                                                |
| $e_t, \bar{e}_t$                      | prequential error at step $t$ , binary; and its inter-seed mean, with no temporal smoothing                                                    |
| $\hat{p}_0$                           | pre-drift error baseline, estimated on the recorded pre-drift steps                                                                            |
| $S_t$                                 | detector state, Eq. (2), with $S_{\tau^*} = 0$                                                                                                 |
| $S_{\max}(H)$                         | peak of that path over the horizon; the alarm is $S_{\max}$ crossing $\lambda$                                                                 |
| $A(H)$                                | accumulated excess error over the horizon                                                                                                      |
| $N_t$                                 | number of distinct trees replaced by step $t$                                                                                                  |
| $\tau_{\text{swap}}(q)$               | first instant at which a fraction $q$ of the $M$ trees has been replaced, so $\tau_{\text{ARF}} = \tau_{\text{swap}}(1/M)$                     |
| $\tau_{50\%}$                         | $\tau_{\text{swap}}(50\%)$ , the instant at which half of the forest has been renewed                                                          |
| $\tau_{\text{rec}}$                   | return of $\bar{e}_t$ below $\hat{p}_0 + \rho \Delta e$ with $\rho = 0.25$ , held 20 consecutive steps; also written $\tau_{\text{err}}(\rho)$ |
| $R(\cdot), G(\cdot)$                  | fraction of the adaptation acquired, and of the evidence accumulated, at a given instant                                                       |
| $\tau_b$                              | Kendall's rank correlation with tie correction, Eq. (4)                                                                                        |

Both delays are counted from  $\tau^*$  and take integer values, a fact the Fréchet bounds of Question A rely on. Every measured figure in this report comes from the instrumented campaign of 20 amplitudes  $\times$  100 seeds = 2000 runs,  $H = 2000$ ,  $M = 10$ , documented in `resultats_R2/`; the two exceptions are the drift-free arm of 100 runs read in Section VII and the ten fictitious runs of the counter-example in Section V-D, both signalled where they are used.

*Intervals in brackets:* A share of runs,  $k$  out of  $n$ , is an estimate of a probability  $p$  from a finite sample, and is reported with its Wilson 95% interval,  $[\ell; u]$ : the set of values of  $p$  under which observing  $k$  successes out of  $n$  would not be rejected at the 5% level. Concretely,  $p$  is retained whenever  $k$  lies within 1.96 standard deviations of  $np$ , the standard deviation  $\sqrt{np(1-p)}$  being taken under  $p$  itself and not under the estimate  $k/n$ ; solving that inequality in  $p$  gives the two bounds. Two consequences matter for reading the tables. The interval never collapses at the edges — 0 out of 100 gives  $[0.00; 0.04]$ , not  $[0; 0]$ , since a rare event can go unobserved in 100 trials — and two shares whose intervals overlap substantially cannot be told apart, whereas disjoint intervals mark a genuine difference. With  $n = 100$  the half-width is about  $\pm 0.10$  near  $p = 0.5$  and narrows towards 0 and 1.

## I. THE MISS PROBABILITY: DEFINITION, IDENTIFICATION, BOUNDS

We formalize and bound  $P_{\text{miss}} = P(\tau_{\text{ARF}} < \tau_{\text{det}})$  without any assumption on the dependence between the two clocks.

### A. Defining the Miss Event

Everything that follows is a statement about  $P_{\text{miss}}$ , and none of it means anything until Miss is unambiguous. The natural reading — the forest repairs itself before the alarm sounds — leaves three cases open: the two clocks coincide, the detector never fires, or neither mechanism ever acts. We settle them by one definition rather than three rules.

a) *A strict inequality:* Both clocks count integer steps, so they can land on the same one and  $P(\tau_{\text{ARF}} = \tau_{\text{det}})$  can be strictly positive: the choice is not cosmetic, and no continuity argument disposes of it — real-valued times can coincide too, all the more so for two functionals of the same error stream. It must therefore be settled on physical grounds. A tie means the alarm sounds *exactly* when the forest repairs itself: the operator is informed at that instant and nothing is lost, unlike the blind spot proper where the alarm comes late or never. A tie is a detection success, whence

$$\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\}$$

on the extended reals, with strict inequality. The convention carries almost no weight in practice: ties occur 0 times out of 1 906 observed races at  $\lambda = 8$ , 6 out of 778 at  $\lambda = 25$ , 0 out of 1 at  $\lambda = 50$ , so the non-strict reading would move  $P_{\text{miss}}$  by at most 0.003. They are rare not for any reason of continuity but because the two mechanisms operate on visibly different time scales — which is the substance of the paradox rather than a technicality.

b) *A silent repair is a miss:* On  $\{\tau_{\text{ARF}} < +\infty, \tau_{\text{det}} = +\infty\}$  the forest repaired itself and the alarm never sounded — the blind spot in its purest form, worse than  $\tau_{\text{ARF}} < \tau_{\text{det}} < +\infty$  where the operator is at least informed late. A finite  $\tau_{\text{ARF}}$  is smaller than  $+\infty$ , so the event falls inside Miss on its own.

It must. Excluding it would restrict the count to runs where the detector eventually spoke — the very selection bias this question holds against the manuscript, since those discarded are the runs where it failed most completely. At  $\lambda = 50$ , where 1 999 runs out of 2 000 never fire, the rate would be computed over a single run.

c) *A double starvation is not:* On  $\{\tau_{\text{ARF}} = +\infty, \tau_{\text{det}} = +\infty\}$  neither mechanism ever acted. The paradox rests on a *silent repair* erasing the evidence; if the repair never happened there is nothing to erase, and the model simply stays broken. Since  $\infty < \infty$  is false, the event is excluded on its own. Its empirical frequency is 0 at all three thresholds, so the convention is never exercised on this grid — but nothing in the model forbids it, and a slower forest would produce it.

### Adopted convention.

$$\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\} \quad (\text{strict, extended reals})$$

A tie goes to the detector, a silent repair is a miss, a double starvation is not — all three from the definition, none from a rule of its own. That is what makes it trustworthy rather than fitted to the answer.

### B. Making the Miss Probability Observable

Miss is defined on the complete race, with no time limit; the campaign observes  $H = 2000$  steps. On runs where neither clock has fired by then the winner is not known, and  $P_{\text{miss}}$  depends on what was never seen: it is not, as it stands, computable from the data. Two ways out are unacceptable — assuming that censored runs behave like the others cannot be checked, and discarding them is the selection bias just rejected. The aim is therefore to replace  $P_{\text{miss}}$  by the probability of an event that *is* observable, or failing that to trap it between two such probabilities.

a) *What the horizon lets us see:* Absolute censoring is  $C_H := \{\min(\tau_{\text{ARF}}, \tau_{\text{det}}) > H\}$ , on which the race is undecided. Sorting runs by the two observable events  $\{\tau_{\text{ARF}} \leq H\}$  and  $\{\tau_{\text{det}} \leq H\}$  gives a partition where the verdict is known everywhere but one cell: regions 1 ( $\leq H$ ,  $\leq H$ , decided directly) and 2 ( $\leq H$ ,  $> H$ , a certain Miss) together form exactly  $\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}$ ; region 3 ( $> H$ ,  $\leq H$ ) is never a Miss; region 4 is  $C_H$ , undetermined.

b) *Trapping the unknown between two computable numbers:* Region 4 can shift the total by its own mass and no more. Write  $L := P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss})$ , computable from the data with no assumption about what happens beyond  $H$ . Counting region 4 as never a Miss gives the lower bound, entirely as Miss the upper one:

$$L \leq P_{\text{miss}} \leq L + P(C_H).$$

This is a Manski-type bound: nothing is assumed about the unobserved runs, their possible influence is bounded by how many there are, and the width of the interval is exactly  $P(C_H)$  — the extent of our ignorance.

c) *On this campaign, the interval collapses:*  $\tau_{\text{ARF}}$  is never censored: 0 runs out of 2 000, the largest value observed being 1 293 steps. The forest, at maximal reactivity ( $c_{\text{int}} = 1$ ), always replaces a tree before the horizon, so  $\min(\tau_{\text{ARF}}, \tau_{\text{det}})$  is always finite,  $C_H = \emptyset$ , and the two bounds coincide:

$$P_{\text{miss}} = P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}).$$

**Result.** The unconditional miss probability equals the probability of an event observable within the horizon. This is an equality, not an assumption: the naive computation would have returned the same number, but nothing would have told us it was right. What the construction adds is the guarantee — and the condition under which it would fail. Were the forest able to stay inert past  $H$ ,  $P_{\text{miss}}$  would cease to be identified and only the interval would survive.

### C. Universal Fréchet Bound

We now drop the finite horizon and ask what the two marginal laws concede on their own, with no assumption on the coupling. This is a different kind of ignorance from the one just treated: there, data were missing; here, a structure is.

a) *Lower bound:* For a fixed  $s \in \mathbb{N}$ , the event  $\{\tau_{\text{ARF}} \leq s < \tau_{\text{det}}\}$  implies  $\tau_{\text{ARF}} < \tau_{\text{det}}$ , so it is a *subset* of Miss. Write  $A = \{\tau_{\text{ARF}} \leq s\}$  and  $B = \{\tau_{\text{det}} > s\}$ , so  $P(A) = F_A(s)$  and  $P(B) = 1 - F_D(s)$ . Fréchet’s inequality follows from  $P(A \cup B) \leq 1$ :

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1,$$

and since a probability is non-negative,  $P(A \cap B) \geq \max(0, P(A) + P(B) - 1)$ . Substituting,  $P(A) + P(B) - 1 = F_A(s) - F_D(s)$ . The inclusion holding for every  $s$ , we keep the largest value:

$$P_{\text{miss}} \geq \max\left(0, \sup_{s \in \mathbb{N}} [F_A(s) - F_D(s)]\right).$$

b) *Upper bound:* Here we proceed by *superset*. We claim

$$\text{Miss} \subseteq \{\tau_{\text{ARF}} \leq s\} \cup \{\tau_{\text{det}} > s + 1\},$$

and prove it by taking a run in Miss and splitting on where  $\tau_{\text{ARF}}$  falls. If  $\tau_{\text{ARF}} \leq s$  the run lies in the first set and nothing more is needed. If  $\tau_{\text{ARF}} \geq s + 1$ , then  $\tau_{\text{det}} > \tau_{\text{ARF}} \geq s + 1$  puts it in the second. In words: either the forest was early, or the alarm — which comes after it — was late. Boole’s subadditivity, then optimizing over  $s$  and truncating at 1, gives

$$P_{\text{miss}} \leq \min\left(1, \inf_{s \in \mathbb{N}} [F_A(s) + 1 - F_D(s + 1)]\right).$$

**Result.** The two bounds above hold for *any* dependence structure between  $\tau_{\text{ARF}}$  and  $\tau_{\text{det}}$ : they use the marginal laws alone, and assume nothing about how the two clocks are coupled. That is what makes them *universal*.

c) *Verification on a toy case:* Take both clocks uniform on  $\{1, 2\}$ , so  $F_A = F_D$  with  $F_A(1) = \frac{1}{2}$ . Then Miss requires  $\tau_{\text{ARF}} = 1$  and  $\tau_{\text{det}} = 2$ , and its probability depends entirely on the coupling: 0 when the two clocks always agree, 0.25 under independence, 0.5 when one is 1 exactly as the other is 2. Identical marginals, three answers. The formulas give a lower bound of 0 and an upper bound of 0.5 (attained at  $s = 0$  and  $s = 1$ ), both reached: the interval is exactly the set of achievable values, and nothing sharper can be had from marginals alone. Assuming independence, as the manuscript does, is therefore not a harmless convenience — it selects one point out of  $[0, 0.5]$ .

d) *On the campaign’s marginals:* Within a cell,  $\hat{F}_A(s)$  is the share of the 100 seeds with  $\tau_{\text{ARF}} \leq s$  and  $\hat{F}_D(s)$  the share with  $\tau_{\text{det}} \leq s$ ; the two bounds are the formulas above evaluated on these.  $\hat{F}_D$  is sub-stochastic — a censored  $\tau_{\text{det}}$  enters it at no finite  $s$  — so censored runs count as detector losses rather than as missing data. At  $\lambda = 8$ :

The bound holds on all 60 cells of the grid (resultats\_R2/scripts/chiffres\_QA.py), and is

Table I  
FRÉCHET BOUNDS ON THE ESTIMATED MARGINALS,  $\lambda = 8$ .

| $\Delta e$ | lower | $\hat{P}_{\text{miss}}$ | upper |
|------------|-------|-------------------------|-------|
| 0.028      | 0.91  | 1.00                    | 1.00  |
| 0.141      | 0.10  | 0.12                    | 0.26  |
| 0.498      | 0.01  | 0.01                    | 0.18  |

Table II  
THE RACE, MEASURED ON THE CAMPAIGN. REGIONS 3 AND 4 ARE EMPTY, SO  $\hat{P}_{\text{miss}}$  ESTIMATES  $P_{\text{miss}}$  ITSELF RATHER THAN A BOUND ON IT.

| $\lambda$ | region 1 | region 2 | $C_H$ | $\hat{P}_{\text{miss}}$ | Wilson 95%       |
|-----------|----------|----------|-------|-------------------------|------------------|
| 8         | 1906     | 94       | 0     | 0.0905                  | [0.0787; 0.1039] |
| 25        | 778      | 1222     | 0     | 0.9715                  | [0.9633; 0.9779] |
| 50        | 1        | 1999     | 0     | 1.0000                  | [0.9981; 1.0000] |

not vacuous: mean width 0.195 at  $\lambda = 8$  and 0.106 at  $\lambda = 25$ . At  $\lambda = 50$  it falls to 0.001, but only because the detector never alarms there,  $\hat{F}_D$  stays flat at 0, and both formulas reach  $P_{\text{miss}} = 1$  on their own — exactness by degeneracy, not by sharpness. At the weakest amplitude the estimated lower bound of 0.91 becomes 0.88 at the 95% level after a bootstrap resampling the 100 seeds. There, the blind spot is forced by the marginals alone, with no assumption on the coupling: it is not an artefact of an independence hypothesis.

### D. The Race, Measured

The event is now defined and its probability identified, so the campaign can be read directly. Table II gives the partition and  $P_{\text{miss}}$  at the three thresholds.

These totals pool cells of very unequal difficulty, and the per-cell breakdown belongs to Question D.

## II. THE INDEPENDENCE ASSUMPTION BETWEEN TREES

The  $M$  trees share the same stream, yet the manuscript treats the  $\tau_i$  as independent when computing  $\tau_{\text{ARF}} = \min_i \tau_i$ . In which direction does that bias its formula?

### A. Conditional Factorization

Once  $S$  is fixed — the entire trajectory  $(x_t, y_t)$  known — what still distinguishes one tree from another is  $U_i$ : the Poisson weights drawn online for bagging, and the random feature subspaces tested at each node. The first is the mechanism the manuscript itself invokes, “Poisson-weighted bootstrap (Oza bagging with  $\lambda = 6$ )” (Section III-G, citing Gomes et al. 2017); the second is a defining property of the Adaptive Random Forest in the same reference<sup>1</sup>.

These draws are independent across trees by construction: tree  $i$  never sees tree  $j$ ’s weights, and a tree created at replacement inherits neither weights nor structure from the one it replaces. Since  $\tau_i$  is a deterministic function of  $(S, U_i)$ , fixing  $S$  leaves only the i.i.d. variation of  $U_i$ . Conditionally on

<sup>1</sup>On the R2 stream it contributes little: with two features only, a random subspace selects roughly one of two at each node. Diversity there rests almost entirely on the Poisson weights, which suffices — only the i.i.d. character of the  $U_i$  is needed.

$S$ , the  $\tau_i$  are therefore independent and identically distributed, and

$$P(\tau_{\text{ARF}} > s \mid S) = \prod_{i=1}^M P(\tau_i > s \mid S) = G_S(s)^M.$$

The independence assumption is thus not wrong in kind — it is stated at the wrong level. It holds given the stream; it fails once the stream is averaged out.

### B. The Direction of the Bias

Unconditioning by the law of total probability gives  $P(\tau_{\text{ARF}} > s) = \mathbb{E}_S[G_S(s)^M]$ . Since  $x \mapsto x^M$  is convex on  $[0, 1]$ , Jensen's inequality yields

$$\mathbb{E}_S[G_S(s)^M] \geq (\mathbb{E}_S[G_S(s)])^M = (1 - F(s))^M,$$

the last equality because  $\mathbb{E}_S[G_S(s)] = P(\tau_1 > s) = 1 - F(s)$ .

#### Result.

$$P(\tau_{\text{ARF}} \leq s) \leq 1 - (1 - F(s))^M$$

The probability of having already adapted by time  $s$  is at most what the independence formula predicts: the manuscript **overestimates the repair speed**.

a) *Equality condition*: Jensen is an equality exactly when  $G_S(s)$  is almost surely constant — when the difficulty of adapting does not depend on which stream was drawn. That never happens here: each seed produces a different noise trajectory after the drift. Independence is therefore never exact, only an approximation, and a systematically optimistic one.

### C. Operational Consequence

Two quantities are affected in opposite directions, and the words below refer to different subjects — which is what makes the pair look contradictory. The independence formula predicts a *faster* repair than the real one, and therefore a *larger* blind-spot risk: a slower repair leaves the monitor more time, so the forest wins the race less often than claimed. Read alone, our correction makes the blind spot *less* severe than the article reports.

a) *Consequence for  $M_{\text{crit}}$* : Corollary 10 of the manuscript asks how many trees the forest may carry before the monitor stops being reliable. Requiring  $P_{\text{miss}} = 1 - [1 - F(\tau_{\text{det}}^*)]^M \leq 1 - \beta$  gives  $[1 - F]^M \geq \beta$ , hence  $M \ln(1 - F) \geq \ln \beta$ ; since  $\ln(1 - F) < 0$  the division flips the inequality:

$$M \leq M_{\text{crit}} := \left\lfloor \frac{\ln \beta}{\ln[1 - F(\tau_{\text{det}}^*)]} \right\rfloor.$$

$M_{\text{crit}}$  is a *ceiling* on the number of trees, not a floor — the physically sensible reading, since more trees mean a faster repair and a worse position for the detector. As the true CDF lies below the independence one, the true  $P_{\text{miss}}$  is smaller at every  $M$ : whatever ensemble size the certificate admits, the real forest satisfies the guarantee too. The corollary is not invalidated but **reclassified as a conservative ceiling**.

b) *Two reservations*: The comparison holds at a *fixed*  $\tau_{\text{det}}^*$ , which Proposition 9 makes a constant. Read as a claim about the real blind-spot risk it assumes more, since the real  $\tau_{\text{det}}$  is computed on the ensemble's error stream and so depends on how correlated the trees are. Both effects in fact lower  $P_{\text{miss}}$  — a slower repair alarms earlier, and weaker variance reduction leaves a burstier stream that breaches  $\lambda$  on noise alone — so the sign holds, but part of the decrease is bought with alarms that win the race without detecting anything.

And the corollary is not rescued. At  $\Delta e = 0.33$  the manuscript measures  $\hat{F}(\tau_{\text{det}}^*) = 0.49$ : one tree alone already gives  $P_{\text{miss}} = 0.49$ , ten times the 5% that  $\beta = 0.95$  allows, whence  $M_{\text{crit}} = 0$ . Our correction states the true ceiling is  $\geq 0$ , which says nothing — rescuing the certificate would require  $\geq 1$ . The status changes, the verdict does not.

### D. The Correlation Trap

Could the dependence run the other way and reverse the conclusion? On the toy case of two trees uniform on  $\{1, 2\}$ , an anti-monotone coupling ( $\tau_1 = 1 \iff \tau_2 = 2$ ) gives  $P(\tau_{\text{ARF}} \leq 1) = 1$  against 0.75 under independence — a *faster* adaptation. Its covariance is  $0 - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4}$ , strictly negative.

No ARF can realize it. By the law of total covariance, with  $X = \mathbf{1}\{\tau_1 > 1\}$  and  $Y = \mathbf{1}\{\tau_2 > 1\}$ , the conditional term vanishes by Section II-A and both conditional expectations equal  $G_S(1)$ , so

$$\text{Cov}(X, Y) = \text{Var}(G_S(1)) \geq 0.$$

**The trap defused.** The only channel of dependence between two trees is the stream they share, and sharing a stream can only make them fail together — never in opposition. The covariance is therefore never negative, and the anti-monotone coupling is structurally out of reach. This is not a missing link in Section II-B, whose proof never invokes the sign of the dependence: it is a second consequence of the same conditional factorization, and it closes the one objection a reader would raise against it.

## III. WHAT THE REPAIR DATE MEASURES, AND THE EVIDENCE IT LEAVES

The manuscript dates the forest's repair at  $\tau_{\text{ARF}}$ , the first tree replacement, and reasons about the race through the evidence the detector accumulates in the meantime. This section examines both objects: what  $\tau_{\text{ARF}}$  dates (Section III-A), how the alarm depends on the error path (Section III-B), and how much evidence the repair leaves behind (Section III-C). The baseline  $\hat{p}_0$  is estimated over the last 3 000 pre-drift steps.

### A. The First Replacement Is a Lower Bound

Let  $N_t$  be the number of distinct trees replaced since  $\tau^*$ , and

$$\tau_{\text{swap}}(q) = \inf\{t > \tau^* : N_t/M \geq q\} - \tau^*.$$

$N_t$  is an integer and never decreases, so  $\tau_{\text{swap}}$  is non-decreasing in  $q$  and  $\tau_{\text{ARF}} = \tau_{\text{swap}}(1/M)$ . Hence, for every  $q \geq 1/M$ ,

$$\tau_{\text{ARF}} \leq \tau_{\text{swap}}(q) \quad \text{on every run.} \quad (1)$$

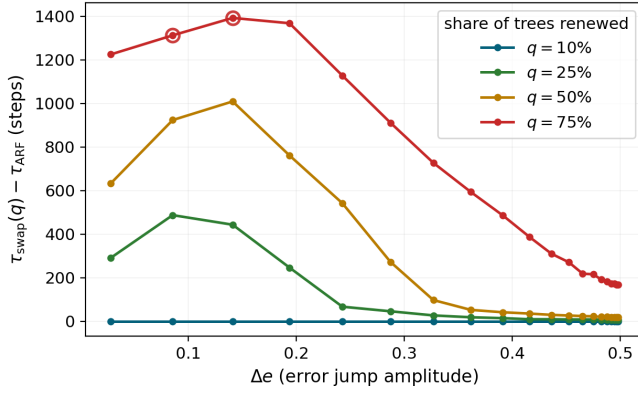


Figure 1. Median gap  $\tau_{\text{swap}}(q) - \tau_{\text{ARF}}$  against  $\Delta e$ , 2000 runs. Hollow markers: more than 50% of the runs censored for that quota ( $q = 75\%$  at  $\Delta e = 0.085$  and  $0.141$ ). The  $q = 10\%$  curve is zero by construction.

With  $M = 10$  the quotas 25% and 75% are first met at 3 and 8 trees; the labels are kept and read as “at least 3 (resp. 8) trees”.

a) *A check, not a finding:* Being deterministic, (1) cannot fail on the forest, only on the instrumentation. The campaign returns  $\tau_{\text{ARF}} = \tau_{\text{swap}}(10\%)$  on all 2000 runs and no violation among the 5795 uncensored comparisons for  $q \in \{25\%, 50\%, 75\%\}$ : the offline reconstruction is sound. What remains to be measured is the size of the gap.

b) *The size of the gap:* Figure 1 gives the median gap by amplitude. Declaring a gap above 100 steps (5% of  $H$ ) disqualifying — a convention fixed before the results were read, whose verdicts do not move anywhere in  $[50, 167]$  —  $\tau_{\text{ARF}}$  is an acceptable substitute for  $\tau_{\text{swap}}(25\%)$  from  $\Delta e \approx 0.24$ , for  $\tau_{\text{swap}}(50\%)$  from  $\Delta e \approx 0.33$ , and never for  $\tau_{\text{swap}}(75\%)$ . Over the 18 amplitudes where fewer than half of the runs are censored, three quarters of the forest is renewed 167 to 1368 steps later, that is 6.0 to 14.3 times  $\tau_{\text{ARF}}$  itself. The bias holds run by run and always on the same side, so averaging over seeds does not reduce it.

c) *At the bottom of the grid, replacements are noise:* With no drift at all, the forest still renews a median of 9 trees out of 10 within the horizon, on 14 swap events per run; at  $\Delta e = 0.028$  the campaign records 12.5. There the quota fills by accident, and the two lowest amplitudes carry a different object from the rest of the grid.

d) *Error recovery comes after the first replacement:* Let  $\tau_{\text{rec}}$  be the return of the seed-averaged error below  $\hat{p}_0 + \rho \Delta e$ ,  $\rho = 0.25$ , held for 20 consecutive steps. It exists on the mean curve only, so it is compared with the median  $\tau_{\text{ARF}}$ , amplitude by amplitude. One could conjecture the reverse order at low amplitude, the surviving trees absorbing a faint shock with no replacement. Without the persistence requirement, four amplitudes violate  $\tau_{\text{ARF}} \leq \tau_{\text{rec}}$ . At the two lowest the test has no power ( $\rho \Delta e < 2\sigma$ ) and the crossing occurs at the very first step. At  $\Delta e = 0.141$  and  $0.194$  it has power, but the crossing, at step 101, is not sustained, and it moves to steps 626 and 322 once persistence is required. No violation remains on the

18 amplitudes where the test has power. Where it is blind the conjecture is not refuted either: resolving it would take about 1900 seeds at  $\Delta e = 0.028$ .

e) *Not the bias of Section II-B:* Both results find  $\tau_{\text{ARF}}$  optimistic, and share nothing else. (1) is combinatorial and would hold unchanged with independent trees; Section II-B compares two laws of  $\tau_{\text{ARF}}$  and says nothing of  $\tau_{\text{swap}}(q)$ . Grant that correction in full, and  $\tau_{\text{ARF}}$  still dates the recovery too early: the two errors compound.

### B. The Alarm Is a Deterministic Function of the Error Path

$\tau_{\text{ARF}}$  counts observations;  $\lambda$  counts excess errors, since the detector

$$S_t = \max(0, S_{t-1} + e_t - \hat{p}_0 - \delta_P), \quad S_{\tau^*} = 0, \quad (2)$$

risers by nearly one unit per error and barely falls on a success. Comparing the two assumes a constant error rate, which the repair precisely destroys; and the reset to zero discards evidence, so the alarm date depends on the order of the errors, not only on their total. The right object is the peak statistic  $S_{\max}(H) = \max_{t \leq H} S_t$ . Write  $X_t = e_t - \hat{p}_0 - \delta_P$  and  $A(k, j) = \sum_{t=k+1}^j (e_t - \hat{p}_0)$ .

**Proposition 1** (Lindley’s closed form). *For every  $j \leq H$ ,  $S_j = \max_{0 \leq k \leq j} \sum_{t=k+1}^j X_t$ , hence*

$$S_{\max}(H) = \max_{0 \leq k \leq j \leq H} [A(k, j) - (j - k) \delta_P].$$

*Proof.* By induction on  $j$ . At  $j = 0$  both sides are 0. If the identity holds at  $j - 1$ , then  $S_j = \max(0, \max_{k \leq j-1} \sum_{t=k+1}^{j-1} X_t + X_j) = \max(0, \max_{k \leq j-1} \sum_{t=k+1}^j X_t)$ , and the 0 is the empty sum obtained at  $k = j$ . Expanding the sum gives the second form.  $\square$

On the 2000 campaign trajectories the closed form matches the recurrence to  $1.26 \times 10^{-11}$ ; the analysis script stops if this check fails.

**Corollary 2** (Certificate).  *$S_{\max}(H) \geq \lambda$  if and only if some sub-window  $]k, j] \subseteq ]0, H]$  satisfies  $A(k, j) \geq \lambda + (j - k) \delta_P$ . This holds on every run, with no averaging and no distributional assumption.*

The full window alone,  $A(0, H) \geq \lambda + H \delta_P$ , is one term of the maximum, and therefore only sufficient.

a) *The full window proves little:* Its toll  $H \delta_P = 20$  turns the thresholds into raw areas of 28, 45 and 70. What a run has to offer against these is the evidence budget of Section III-C: about 18 units whatever the amplitude, as the manuscript’s rectangle and its fit predict (Sections III-C and III-D there). That figure is an expectation, what a typical run supplies, not a ceiling on any single run; all three requirements exceed it, so only runs well above the average can meet them, and the higher the bar the fewer they are.  $A(0, H)$  reaches 28 in 473 of the 2000 runs, 45 in 66, and 70 in none. The full window thus certifies detection for 23.7% of runs at  $\lambda = 8$  (Wilson 95% interval  $[21.8; 25.6]$ ), 3.3% at  $\lambda = 25$   $[2.6; 4.2]$  and none at  $\lambda = 50$ . Failing a sufficient condition proves nothing: the full window is simply the wrong place to look.

Table III

SHORT-WINDOW CERTIFICATE  $S_{\max}(w) \geq \lambda$  AGAINST DETECTION WITHIN  $H$ , AS A SHARE OF THE 100 SEEDS OF EACH AMPLITUDE; BRACKETS: WILSON 95% INTERVAL. AT  $\lambda = 50$  THE CERTIFICATE IS 0.00 AT EVERY AMPLITUDE.

| $\Delta e$ | $w$  | $\lambda = 8$              |                   | $\lambda = 25$             |                   |
|------------|------|----------------------------|-------------------|----------------------------|-------------------|
|            |      | $S_{\max}(w) \geq \lambda$ | alarm by $H$      | $S_{\max}(w) \geq \lambda$ | alarm by $H$      |
| 0.028      | 1969 | 0.10 [0.06; 0.17]          | 0.10 [0.06; 0.17] | 0.00 [0.00; 0.04]          | 0.00 [0.00; 0.04] |
| 0.085      | 650  | 0.95 [0.89; 0.98]          | 0.96 [0.90; 0.98] | 0.09 [0.05; 0.16]          | 0.15 [0.09; 0.23] |
| 0.141      | 394  | 1.00 [0.96; 1.00]          | 1.00 [0.96; 1.00] | 0.44 [0.35; 0.54]          | 0.70 [0.60; 0.78] |
| 0.243      | 229  | 1.00 [0.96; 1.00]          | 1.00 [0.96; 1.00] | 0.78 [0.69; 0.85]          | 0.94 [0.88; 0.97] |
| 0.498      | 112  | 1.00 [0.96; 1.00]          | 1.00 [0.96; 1.00] | 0.02 [0.01; 0.07]          | 0.02 [0.01; 0.07] |

*b) The short window:* Anchoring instead on  $w = \text{round}(3 \times 18.5/\Delta e)$  steps after the drift and checking  $S_{\max}(w) \geq \lambda$  gives Table III. The window being fixed in advance and shorter than  $H$ , the certificate is sufficient for detection within  $H$  but not necessary — an alarm may fire after  $w$  — so it can never exceed observed detection: what informs is their gap. At  $\lambda = 8$  it vanishes (mean 0.0005 over the 20 amplitudes): the alarm fires inside the transient. At  $\lambda = 25$  it reaches 0.26 at  $\Delta e = 0.141$ : those alarms fire after the transient and cannot be credited to the drift window. At  $\lambda = 50$  the certificate is 0 at every amplitude.

### C. An Almost Constant Evidence Budget

The manuscript’s stylized transient is a rectangle: an excess error  $\Delta e$  over a width  $\tau_{\text{ARF}}$ . Combined with its fit  $\mathbb{E}[\tau_{\text{ARF}}] \approx 18.5 \Delta e^{-0.98}$ , this predicts

$$\mathbb{E}[A] \approx \Delta e \cdot \mathbb{E}[\tau_{\text{ARF}}] \approx 18.5 \Delta e^{0.02}, \quad (3)$$

which varies by 5.9%, from 17.2 to 18.2, while the amplitude varies by a factor of 17.7: the forest repairs all the faster as the shock is violent, and a violent shock leaves the detector no more evidence than a slow drift. The tolerance makes this worse where it matters.  $\delta_P$  removes a share  $\delta_P/\Delta e$  of the budget, 35.5% at  $\Delta e = 0.028$  against 2.0% at 0.498, leaving a usable budget of 11.1 to 17.9. None of the full-window requirements of 28, 45 or 70 is within reach on average, and observing longer widens the deficit, since the toll grows linearly in  $H$  while the budget does not.

*a) The measurement confirms the fact, not the derivation:* Over  $\Delta e \in [0.085; 0.498]$ , where the amplitude varies by a factor of 5.8, the measured short-window area  $A(w)$  varies by 1.78 (Figure 2): the compensation is real, though weaker than the 1.04 predicted, and the model underestimates the area mid-grid, 30.8 measured against 18.0 at  $\Delta e = 0.243$ . Including  $\Delta e = 0.028$ , where the short window nearly covers the horizon and the signal sits below the noise floor, the factor becomes 11.4, so the restriction is part of the result. Both hypotheses stacked in (3) fail. The fit holds only mid-grid: measured over predicted  $\tau_{\text{ARF}}$  is 0.19 at  $\Delta e = 0.028$ , 2.08 at 0.141, 1.02 at 0.287 and 0.76 at the top. The rectangle, recomputed from the measured  $\tau_{\text{ARF}}$ , overestimates  $A(w)$  by 23% to 43% below  $\Delta e = 0.19$  and underestimates it by 6.9% to 41.1% above, changing sign in between:  $\tau_{\text{ARF}}$  is not the width of the transient, as Section III-A predicts. At large amplitude a third failure

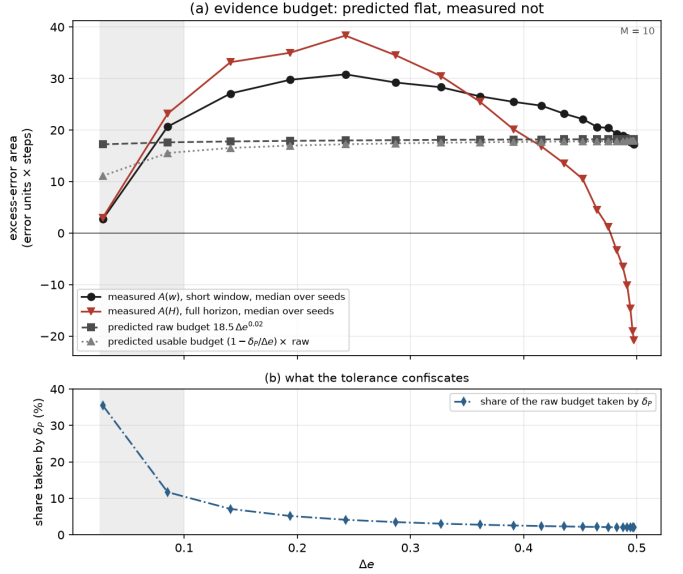


Figure 2. (a) Predicted budgets, raw  $18.5 \Delta e^{0.02}$  and usable, against the measured areas  $A(w)$  and  $A(H)$ , medians over 100 seeds. (b) Share of the raw budget taken by the tolerance  $\delta_P$ . Shaded: the two amplitudes outside the decision domain of Question D.

appears. The post-drift task is easier than the original one, the error settles below its baseline — on 1934 of the 2000 steps at  $\Delta e = 0.498$  — and  $A(H)$  turns negative from  $\Delta e = 0.482$ , down to  $-20.8$ .

**Result.** The evidence the repair leaves to the detector is nearly independent of the drift’s amplitude — a factor of 1.8 against 5.8 — and the tolerance erodes it most at low amplitude. This rests on direct measurement: the manuscript’s derivation, its exponent 0.02 and its rectangle, is scaffolding that the measurement replaces rather than supports.

### IV. INSTRUMENTING THE REPAIR, THE ERROR AND THE DETECTOR

The subject asks for the fraction of replaced trees, the global error and the internal state of the detector to be traced jointly. This section produces those three series and the four temporal markers read off them; what they are worth as measurements of the adaptation is settled in Sections V and VI. Every figure quoted is repro-

duced by `resultats_R2/scripts/chiffres_QD.py` and `derives_QD.py`.

#### A. What Is Recorded and What Is Recomputed

The simulation writes raw material only: the error trajectory  $e_t \in \{0, 1\}$ , the tree replacement events, and the 3 000 pre-drift steps. No threshold and no  $\lambda$  appears anywhere in the simulation loop.

That is deliberate. The external detector reads the error trajectory and never acts on the forest, so  $\lambda$  has no influence on the dynamics. Once the `break` is removed, the three scenarios of R2 produce runs identical to the bit. One campaign replaces three, and  $S_t$ ,  $\tau_{\text{det}}$ ,  $A(H)$  and  $S_{\text{max}}(H)$  are swept offline for every threshold and any  $\delta_P$ . Freezing a threshold inside the loop would have cost one campaign per threshold, and at  $\lambda = 50$  it would have produced a table whose  $\tau_{\text{det}}$  column is censored in 99.95% of runs. It is also what makes the continuous threshold sweep of Section VII cost nothing.

Two conventions are used throughout and stated once here. The detector is created at  $\tau^*$  with  $S_{\tau^*} = 0$ , and every duration is counted from  $\tau^*$ . The two amplitudes below  $\Delta e = 0.10$  lie outside the decision domain fixed in Section VI-B and are shaded on every figure that sweeps  $\Delta e$ .

The hierarchy (1) of Section III-A is used throughout, and with it the identity  $\tau_{\text{ARF}} = \tau_{\text{swap}}(1/M)$ : that is what makes the first row of Table IX tautological rather than measured.

#### B. The Three Series on a Common Time Axis

a) *No temporal smoothing*:  $e_t$  is binary. The only averaging applied is the inter-seed mean, taken step by step, with no sliding filter anywhere. A moving average of length  $L$  shifts the curve by half its support and destroys the synchronisation with  $\tau^*$ , which is what this figure exists to display. At the top of the grid the median  $\tau_{\text{ARF}}$  is 28 steps, so a 50-step filter would displace the curve by nearly the whole quantity being measured.

b) *The four markers*:  $\tau_{\text{det}}$  is plotted at  $\lambda = 8$ , the only threshold whose censoring stays in the minority (4.70%, against 61.10% at  $\lambda = 25$  and 99.95% at  $\lambda = 50$ ). At the other two thresholds a median delay would rest on a minority of runs, so none is published.  $\tau_{\text{rec}}$  carries the persistence requirement whose necessity Section III-A establishes: four amplitudes cross the threshold without the crossing being sustained, and the marker is dashed where the test has no power.

c) *Replacement dynamics*: At  $t = 100$  the median  $N_t/M$  ranges from 0.00 to 0.70 across amplitudes, with an interquartile range of 0.00 to 0.30; at  $t = 400$ , from 0.10 to 0.90. By the end of the horizon the median runs from 0.60 to 1.00, and the proportion of runs whose forest has been entirely renewed goes from 2% to 94% depending on amplitude.

#### C. The Baseline and Its Estimation Window

The baseline  $\hat{p}_0$  enters  $S_t$  directly, and therefore governs  $\tau_{\text{det}}$ ,  $A(H)$ ,  $S_{\text{max}}(H)$  and every censoring rate reported in this work. It is reported on both windows the protocol requires: 1 000 pre-drift steps, reproducing R2 as published, and 3 000

Table IV  
BASELINE AND ITS CONSEQUENCES ON BOTH ESTIMATION WINDOWS. THE MAXIMUM PER-RUN DIFFERENCE BETWEEN THE TWO BASELINES IS 0.009333.

| Window      | $\hat{p}_0$ (median) | mean censoring of $\tau_{\text{det}}$ |                |                |
|-------------|----------------------|---------------------------------------|----------------|----------------|
|             |                      | $\lambda = 8$                         | $\lambda = 25$ | $\lambda = 50$ |
| 1 000 steps | 0.022000             | 0.0405                                | 0.6045         | 0.9990         |
| 3 000 steps | 0.023000             | 0.0470                                | 0.6110         | 0.9995         |

steps for the analysis. Since 3 000 pre-drift steps are recorded per run, this is a purely offline recomputation. As a control, the baseline recomputed at 3 000 steps reproduces the campaign value with a maximum deviation of exactly 0.

The window shifts the censoring rates by less than one point and changes no verdict anywhere in this report. The direction of the shift is systematic: a baseline estimated over 3 000 steps is slightly higher, so the detector accumulates slightly less evidence and alarms slightly less often. Unless stated otherwise, every figure below uses the 3 000-step baseline.

a) *Against the theoretical baseline*: The pre-drift target  $y = \mathbf{1}\{x_0 + x_1 > 0\}$  is a deterministic function of the inputs. There is no label noise and the Bayes error is exactly 0, so the measured baseline of 0.023 is entirely approximation error, produced by axis-parallel splits approximating an oblique boundary. The detector monitors departures from a reference level that belongs to the classifier rather than to the problem.

#### D. What Is Acquired at $\tau_{\text{ARF}}$

Two quantities are measured at the instant  $\tau_{\text{ARF}}$ : the fraction of adaptation already acquired,  $R(\tau_{\text{ARF}})$ , and the fraction of evidence already accumulated by the detector,  $G(\tau_{\text{ARF}})$ .  $G$  is computed run by run,  $R$  per amplitude on the mean curve, and  $\tau_{\text{ARF}}$  is the inter-seed median.

a) *Answer*:  $G$  ranges from 0.4889 to 0.7594 over the 18 amplitudes of the decision domain: between half and three quarters of the evidence is already available to the detector when the first tree is replaced. On the full grid it falls to 0.1023, and that restriction is published with the figure.

$R$  supports no such single sentence. It runs from  $-0.1882$  to  $+0.8853$  on the grid, from  $-0.1882$  to  $+0.5883$  on the decision domain, and it is not monotone in  $\Delta e$ . At  $\Delta e = 0.085$  it reaches 0.885: nearly all of the adaptation is acquired when the first tree is replaced, the opposite of what one would expect. At  $\Delta e = 0.141$  it is 0.588. The reading “early in the adaptation, late in the race” is not supported by the table and is not used here. What the measurement supports is narrower: at  $\tau_{\text{ARF}}$ , between 49% and 76% of the evidence is accumulated, while the fraction of adaptation acquired varies from negative to 89% with no monotone dependence on amplitude.

b) *A negative fraction of adaptation*:  $R$  is negative at five amplitudes ( $\Delta e = 0.243, 0.436, 0.494, 0.496, 0.498$ ), reaching  $-0.1882$ . The window used to estimate the post-drift plateau is bounded by  $\tau_{\text{ARF}}/2$  precisely to keep it from overlapping the adaptation it measures, and the bound is active at those amplitudes (the window drops to 43, 17, 14, 14, 14 steps

Figure 3. Fraction of replaced trees  $N_t/M$  (inter-seed median and interquartile range), global error  $\bar{e}_t$ , and internal detector state  $S_t$  on a common time axis, at three amplitudes spanning the grid, with the four temporal markers  $\tau_{\text{det}}$  (at  $\lambda = 8$ ),  $\tau_{\text{ARF}}$ ,  $\tau_{50\%}$  and  $\tau_{\text{rec}}$ . The time axis is logarithmic: the four instants span two orders of magnitude across the grid, and on a linear axis the markers of the high-amplitude column collapse into the first few pixels. The  $\tau_{\text{rec}}$  marker is dashed where the test lacks the power to resolve it.

Question D.1 — replacement, error and detector on a common time axis

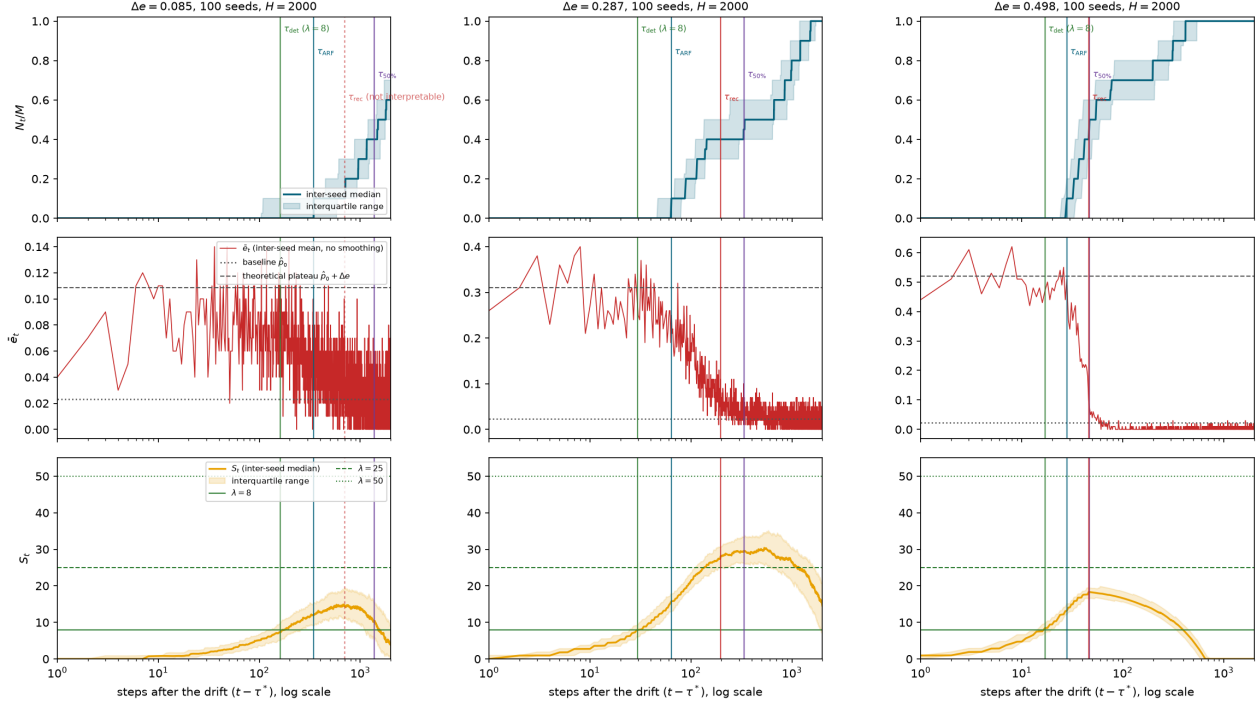
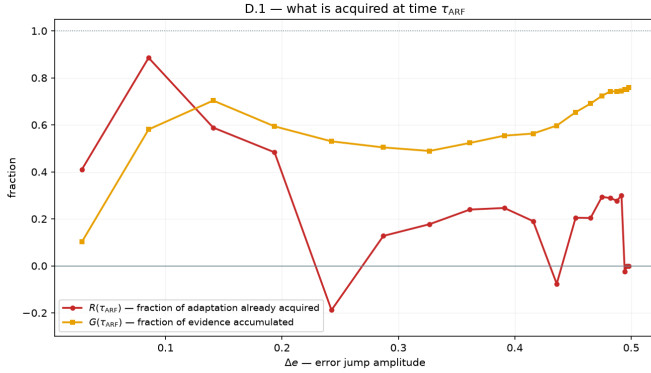


Figure 4.  $R(\tau_{\text{ARF}})$  and  $G(\tau_{\text{ARF}})$  against  $\Delta e$ . Neither is monotone, and  $R$  is negative at five amplitudes.



respectively). The symptom persists anyway. A negative  $R$  means the mean error at  $\tau_{\text{ARF}}$  sits *above* the plateau estimated just after the drift: the error is still rising when the first tree is replaced. Calling the estimation artefact repaired would be false.

c) *The plateau never reaches its theoretical value:* The control of the measured plateau against  $\hat{p}_0 + \Delta e$  fails on all 20 amplitudes, with a deviation between  $-0.0271$  and  $-0.0080$ , always negative. Two mechanisms account for it, and both point at the baseline.

First,  $\hat{p}_0$  is itself partly a product of the replacement

mechanism. On the drift-free arm of the campaign — 100 runs with no change point at all, reported in full in Section VII — the forest renews most of itself over the horizon anyway, for a mean error of 0.0226. Replacement churn under pure noise is the normal regime. A direct comparison against a forest that never replaces anything gave 0.0125, but that figure is not produced by this campaign: it comes from an exploratory measurement of 12 executions (resultsats\_R2/JOURNAL.md § 2 b) whose output was not persisted, and it is quoted as an order of magnitude. Establishing the permanent cost of the replacement mechanism properly is an open item of that journal.

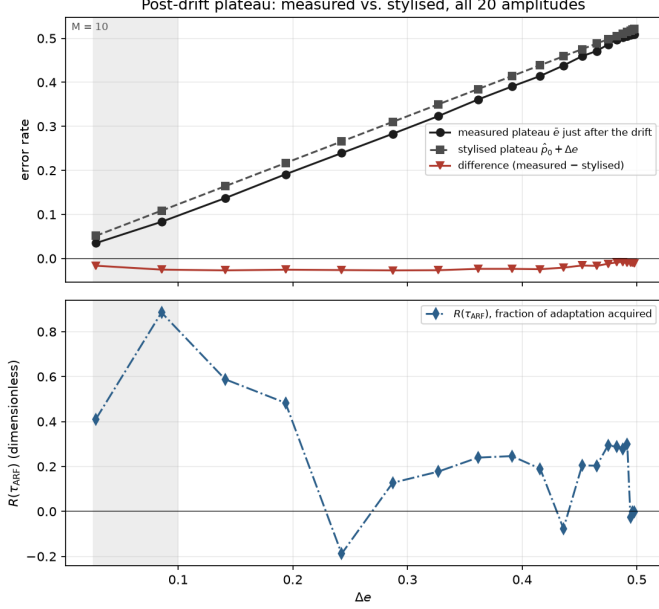
Second, at large amplitudes the post-drift task is easier than the original one, so the error does not settle at  $\hat{p}_0 + \Delta e$  at all — the same mechanism that drives  $A(H)$  negative in Section III-C. The stylised model of the subject fails at the top of the grid, and that is a measurement rather than a conjecture.

### E. The Race, Amplitude by Amplitude

Table II gives  $P_{\text{miss}}$  pooled over the whole grid. The instrumentation makes visible what that pooling hides, and the reading below is the same race under its other orientation: the detector wins exactly when the forest does not.

At  $\lambda = 8$ ,  $\tau_{\text{det}} < \tau_{\text{ARF}}$  in 95.44% of the 1906 uncensored runs, or 90.95% of all 2000 runs when censored ones are imputed as losses. Among those 2000 runs, 34 have  $\tau_{\text{ARF}} = 0$ : a tree is replaced at the change point itself, and the detector loses by construction, since  $\tau_{\text{det}}$  cannot precede the drift. They

Figure 5. Top: the measured post-drift plateau  $\bar{e}$  against the stylised plateau  $\hat{p}_0 + \Delta e$ , and their difference, on all 20 amplitudes; the difference is negative everywhere, between  $-0.0271$  and  $-0.0080$ . Bottom:  $R(\tau_{\text{ARF}})$  with the line at zero, negative at five amplitudes and not monotone. Table QCD\_R\_et\_G\_full.



occur at every amplitude, three at  $\Delta e = 0.028$  down to one at the top of the grid, and Figure 7 plots them explicitly. At  $\lambda = 25$  the proportion falls to 6.56%, and at  $\lambda = 50$  it is 0% on the single uncensored run.

Those grid averages need one qualification, because the rate is far from stationary in  $\Delta e$ . At  $\lambda = 8$  the detector wins 0 of the 10 uncensored runs at  $\Delta e = 0.028$  (where 90% of runs never alarm at all), 64.6% at  $\Delta e = 0.085$  (62 of 96 uncensored runs, Wilson 95% interval 54.6% to 73.4%), and between 88% and 99% everywhere above. The 95.44% figure is dominated by the 18 easy amplitudes out of 20. The blind spot is therefore a property of the pair  $(\lambda, \Delta e)$  rather than of  $\lambda$  alone: even at the most sensitive threshold tested, it is fully present at the bottom of the grid.

Table V gives the race amplitude by amplitude at  $\lambda = 8$ , and Figure 6 the same for the three thresholds together with the censoring that each cell carries. Both are the material Section I reads as  $P_{\text{miss}}$ , resolved in  $(\lambda, \Delta e)$  instead of pooled; Section VII continues the same sweep through the thresholds between 8 and 25, where the transition happens.

## V. CHOOSING THE ASSOCIATION COEFFICIENT

The subject asks for the correlation of  $\tau_{\text{ARF}}$  with the real adaptation. Which coefficient answers that question is not a detail here: the quantities to be correlated are durations measured on a finite horizon, and two of the three usual coefficients are disqualified by that alone. The counter-example of Section V-D comes from `scripts/contre_exemple_kendall.py`, which computes both coefficients twice, once from their definition and once through `scipy`, so that each controls the other.

## A. Three Features of the Data

a) *Non-monotonicity*: The median  $\tau_{\text{ARF}}$  is 118 steps at  $\Delta e = 0.028$ , rises to 346 at  $\Delta e = 0.085$ , then falls steadily to 28 at the top of the grid: a range of 28 to 346, a factor of 12.4, and not monotone in  $\Delta e$ . The rise is not an artefact of the median. The means move the same way (162.7 then 385.9), and the bootstrap confidence intervals of the two medians are disjoint,  $[92.5; 158.5]$  against  $[281; 455]$  (Table VI). The *location* of the peak is less certain: the interval at  $\Delta e = 0.141$ ,  $[231.5; 307]$ , overlaps the one at  $\Delta e = 0.085$  over  $[281; 307]$ , so the two cells are not separated by the rule used here and the claim is that a peak exists at low amplitude, not that it sits at one particular cell. That negative statement is weak, and the control quantifies how weak: at a ratio of medians of 1.32, which is exactly  $346/263$ , the rule separates two samples of 100 in only 23.4% of 500 repetitions on a log-normal law. Failing to separate these two cells is therefore close to uninformative, whereas the separation of  $\Delta e = 0.028$  from  $\Delta e = 0.085$  is not: the rule falsely separates two samples of the same law in 1.0% of repetitions. The relationship is therefore worse than merely curved, which would already understate Pearson's  $r$ . No single linear coefficient summarises it, and its sign depends on which part of the grid is included. Figure 7 shows the full distribution at every amplitude.

b) *Extreme values*: At low amplitude the durations are large and widely dispersed. Pearson's coefficient is a ratio of moments and is not robust, so a handful of runs in the tail move it substantially. Both rank coefficients are unaffected.

c) *Administrative censoring at H*: A duration never reached within the horizon has no numerical value.  $\tau_{\text{swap}}(75\%)$  is censored in 9.10% of runs overall and in up to 78% at a single amplitude;  $\tau_{\text{det}}$  at  $\lambda = 50$  is censored in 99.95%.

d) *Which survives stratification*: Section VI holds  $\Delta e$  constant within a stratum, and the three features do not survive that equally. The first does not survive at all: the non-monotonicity is non-monotonicity in  $\Delta e$ , and within a stratum  $\Delta e$  is fixed by construction. The other two survive, and they are objections of different kinds. Extreme values make Pearson's coefficient inefficient: it can still be computed, it is merely unstable and moves with a few runs. Censoring makes it undefined, because a censored run has no numerical value to enter the sum of products, so the coefficient cannot be formed without first inventing one. The second objection carries the argument, since the first is a matter of degree and the second a matter of definition. It is also the objection that bites on our data, where  $\tau_{\text{swap}}(75\%)$  is censored in nearly four runs out of five at  $\Delta e = 0.085$ .

## B. Why Censored Runs May Be Ranked Last

Both rank coefficients handle censoring the same way and for the same reason. The horizon  $H = 2000$  is identical for every run, so a duration not reached within it genuinely exceeds every observed duration, and censored runs may be placed at the last rank, tied among themselves.

The argument depends entirely on the horizon being common, and it is stated here because every coefficient in this report

Table V

THE RACE AT  $\lambda = 8$ , AMPLITUDE BY AMPLITUDE. THIRD COLUMN: RUNS IN WHICH THE DETECTOR ALARMS STRICTLY BEFORE THE FIRST TREE REPLACEMENT. FOURTH COLUMN: THAT COUNT OVER THE UNCENSORED RUNS OF THE CELL, WITH A WILSON 95% INTERVAL; IT IS A STATISTIC ON SURVIVORS AND IS NOT REPORTED WHERE MORE THAN HALF OF THE CELL IS CENSORED. FIFTH COLUMN: THE SAME COUNT OVER ALL 100 RUNS, CENSORED RUNS COUNTED AS DETECTOR LOSSES. THE LAST ROW IS THE GRID TOTAL,  $1\,819/1\,906 = 95.44\%$  (WILSON 94.4% TO 96.3%) AND  $1\,819/2\,000 = 90.95\%$  (WILSON 89.6% TO 92.1%). THOSE TWO INTERVALS MEASURE SAMPLING ONLY: BOTH TOTALS POOL CELLS OF VERY UNEQUAL DIFFICULTY, AND IT IS THE SPREAD DOWN THE COLUMN, NOT THE INTERVAL, THAT CARRIES THE QUALIFICATION. TABLE QCD\_INDICATEURS\_FULL, COMPUTED BY CHIFFRES\_QD.PY.

| $\Delta e$ | uncensored / 100 | $\tau_{\text{det}} < \tau_{\text{ARF}}$ | share of uncensored, % [Wilson 95%] | share of all 100, % |
|------------|------------------|-----------------------------------------|-------------------------------------|---------------------|
| 0.028      | 10               | 0                                       | not reported (censoring 90%)        | 0.0                 |
| 0.085      | 96               | 62                                      | 64.6 [54.6; 73.4]                   | 62.0                |
| 0.141      | 100              | 88                                      | 88.0 [80.2; 93.0]                   | 88.0                |
| 0.194      | 100              | 94                                      | 94.0 [87.5; 97.2]                   | 94.0                |
| 0.243      | 100              | 97                                      | 97.0 [91.5; 99.0]                   | 97.0                |
| 0.287      | 100              | 97                                      | 97.0 [91.5; 99.0]                   | 97.0                |
| 0.327      | 100              | 97                                      | 97.0 [91.5; 99.0]                   | 97.0                |
| 0.361      | 100              | 98                                      | 98.0 [93.0; 99.4]                   | 98.0                |
| 0.391      | 100              | 98                                      | 98.0 [93.0; 99.4]                   | 98.0                |
| 0.416      | 100              | 98                                      | 98.0 [93.0; 99.4]                   | 98.0                |
| 0.436      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.452      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.465      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.475      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.482      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.488      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.492      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.494      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.496      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| 0.498      | 100              | 99                                      | 99.0 [94.6; 99.8]                   | 99.0                |
| all 20     | 1 906            | 1 819                                   | 95.44                               | 90.95               |

Table VI

MEDIAN OF  $\tau_{\text{ARF}}$  WITH ITS PERCENTILE BOOTSTRAP CONFIDENCE INTERVAL (CI), 10 000 RESAMPLES OF THE 100 SEEDS OF THE AMPLITUDE, GENERATOR SEED 0. THE ROW AT  $\Delta e = 0.141$  IS THERE SO THAT THE OVERLAP DISCUSSED ABOVE CAN BE CHECKED. TABLE QCD\_IC\_MEDIANES\_TAU\_ARF. BEFORE BEING APPLIED TO THE DATA THE ESTIMATOR PASSED FIVE CONTROLS ON CASES WITH A KNOWN ANSWER, STORED IN QCD\_IC\_MEDIANES\_TEMOINS: ON 500 SAMPLES OF SIZE 100 FROM A LOG-NORMAL LAW OF KNOWN MEDIAN, THE INTERVAL COVERED THE TRUE MEDIAN IN 95.2% OF CASES; ON 500 PAIRS OF SAMPLES FROM THE SAME LAW, THE INTERVAL OF THE DIFFERENCE OF MEDIANS CONTAINED 0 IN 96.4% OF CASES; ON A CONSTANT SAMPLE THE INTERVAL COLLAPSED TO THE POINT; AND THE TWO CONTROLS THAT BEAR ON THE RULE ACTUALLY USED HERE, NAMELY DECLARING TWO MEDIANS DISTINCT WHEN THEIR INTERVALS ARE DISJOINT, GAVE A FALSE-SEPARATION RATE OF 1.0% UNDER THE NULL AND A SEPARATION RATE OF 23.4% AT A RATIO OF MEDIANS OF 1.32.

| $\Delta e$ | median $\tau_{\text{ARF}}$ (steps) | 95% CI        | mean  | seeds |
|------------|------------------------------------|---------------|-------|-------|
| 0.028      | 118                                | [92.5; 158.5] | 162.7 | 100   |
| 0.085      | 346                                | [281; 455]    | 385.9 | 100   |
| 0.141      | 263                                | [231.5; 307]  | 331.9 | 100   |
| 0.498      | 28                                 | [27; 29]      | 29.4  | 100   |

relies on it. If  $H$  varied from run to run, a censored value would only be known to exceed *its own* horizon: a run censored at  $H = 500$  and an observed run of 1 200 steps could not be ordered at all, and no imputation to a shared last rank would be licit. Under variable horizons the correct tool is a censoring-aware estimator such as Harrell's  $C$  index. The convention used here rests on a property of this experimental design and does not transfer. Section VI-G checks what it changes in practice: nothing that carries a verdict.

That common horizon is also where the choice between Spearman and Kendall becomes decisive.

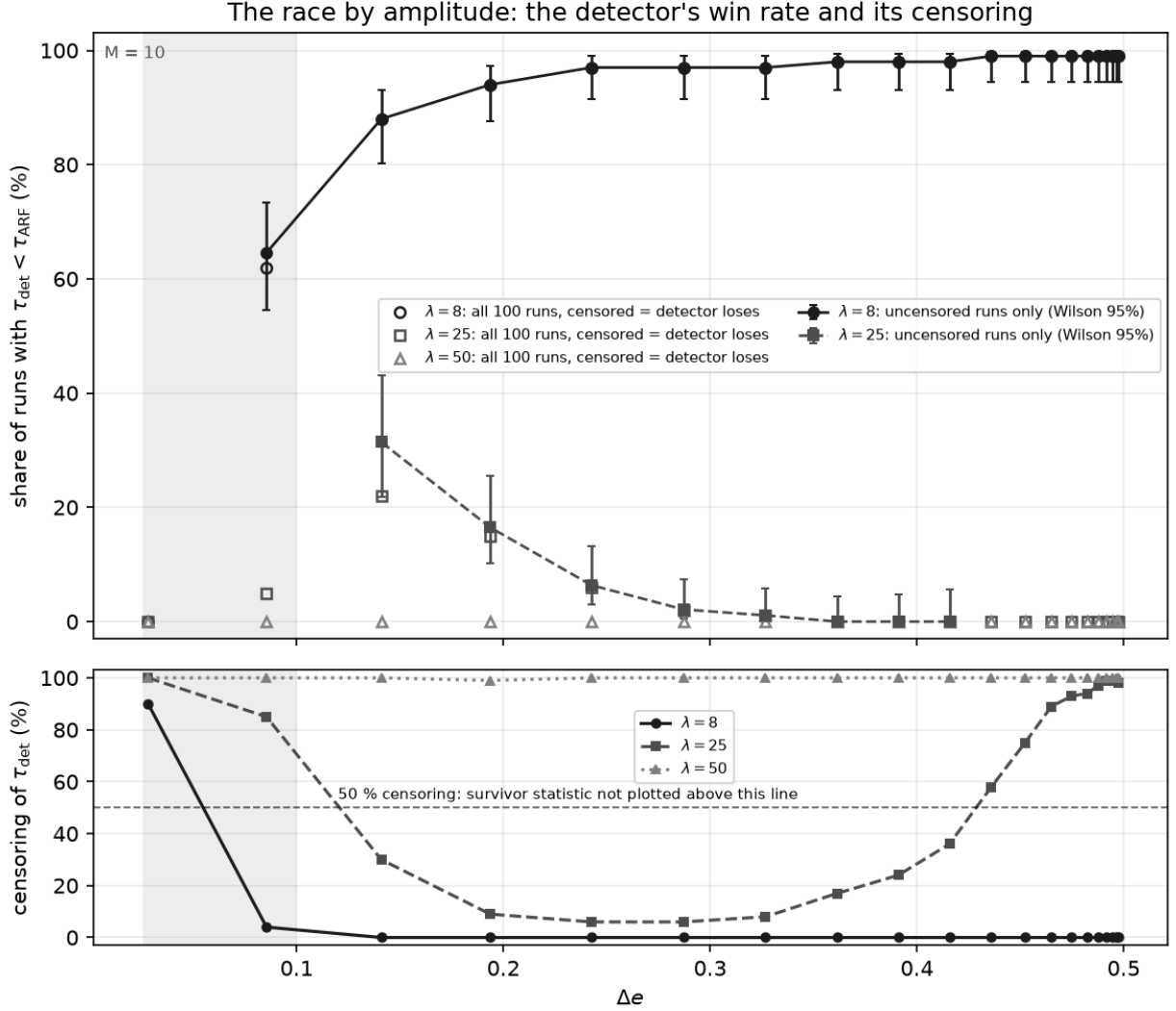
### C. Definition of $\tau_b$ , and Two Tie Structures

For  $n$  pairs, let  $C$  be the number of concordant pairs,  $D$  the discordant ones, and  $n_0 = n(n-1)/2$ . Let  $t_x$  count the pairs tied on  $x$  only and  $t_y$  those tied on  $y$  only, pairs tied on both sides being counted in each. Then

$$\tau_b = \frac{C - D}{\sqrt{(n_0 - t_x)(n_0 - t_y)}}. \quad (4)$$

The denominator distinguishes  $\tau_b$  from the ordinary  $\tau_a$ : each tie block is removed from the count on the side where it occurs, so the coefficient is normalised by the pairs that could in principle have been ordered on both sides. Spearman's  $\rho$  has no such correction. It is Pearson's coefficient computed on mid-ranks,

Figure 6. Top: share of runs in which  $\tau_{\text{det}} < \tau_{\text{ARF}}$ , by amplitude, for  $\lambda \in \{8, 25, 50\}$ . Filled markers with error bars: uncensored runs only, Wilson 95% interval, drawn only where the censoring of the cell is at most 50%. Hollow markers: all 100 runs, censored runs counted as detector losses, drawn everywhere. Bottom: censoring of  $\tau_{\text{det}}$  in the same cells, with the 50% line. What to see: at  $\lambda = 8$  the detector wins almost everywhere except at the bottom of the grid; at  $\lambda = 25$  its win rate falls from 31% at  $\Delta e = 0.141$  to zero by  $\Delta e = 0.36$  while its censoring climbs back to 99% at the top of the grid; at  $\lambda = 50$  it never wins. The fractions plotted are those of Table V, checked for equality inside the script.



and a tie block simply receives a common mid-rank that then enters a sum of squares as an ordinary value.

The comparators used below do not present the same tie structure, and the consequence differs.

a) *Ties on both sides:  $\tau_{\text{ARF}}$  against  $\tau_{\text{swap}}(q)$ .*: Both are durations and both can be censored at  $H$ . Censored runs form a tie block on the  $y$  side.  $\tau_{\text{ARF}}$  is never censored (0.0000), but it is discrete and takes only 321 distinct values over 2 000 runs, so it carries tie blocks of its own. Equation (4) corrects on both sides and the compensation is complete: the coefficient is normalised by exactly the set of orderable pairs.

b) *Ties on one side:  $\tau_{\text{ARF}}$  against  $A(H)$  and  $S_{\text{max}}(H)$ .*: These two are defined on every run and never censored: 2 000 values out of 2 000, with 1 389 and 1 544 distinct values respectively. There is no tie block on their side at all, so the correction applies to the  $x$  side only. The denominator still

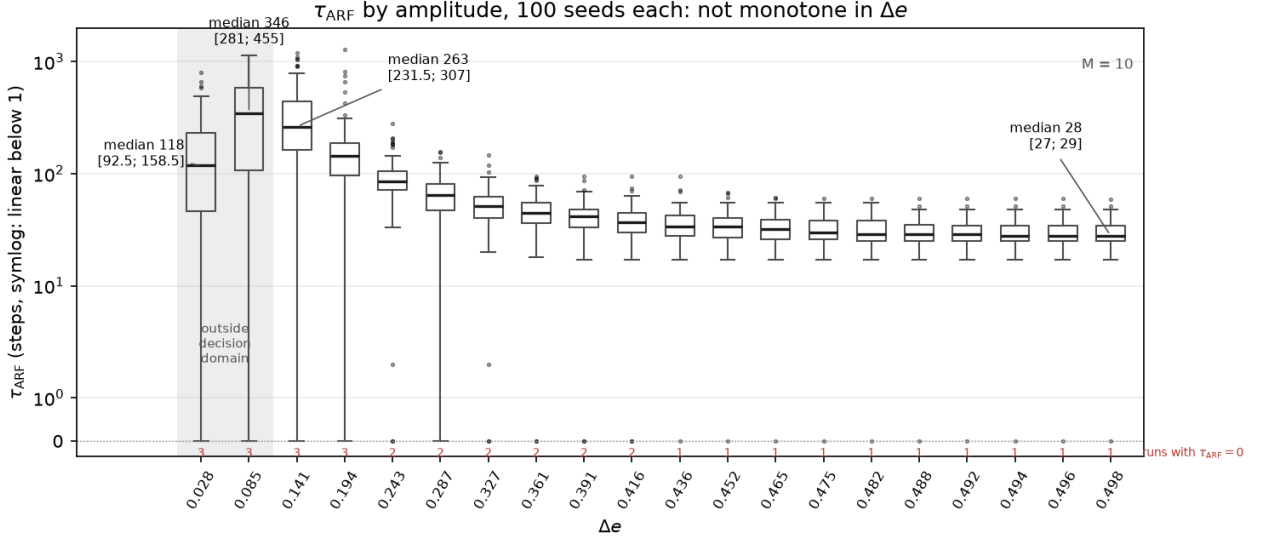
counts the orderable pairs, but the two sides are no longer symmetric: ties in  $\tau_{\text{ARF}}$  remove pairs from the numerator without any counterpart, and the coefficient is displaced *towards zero* relative to a tie-free measurement of the same association.

That direction is what makes the central result of Section VI readable. The coefficients reported there for  $A(H)$  and  $S_{\text{max}}(H)$  are conservative: a value indistinguishable from zero cannot be an artefact of ties inflating it, and a value that is discernible is if anything understated.

#### D. The Counter-Example, and an Error in the Prescribed Exercise

Ten fictitious runs on a common horizon  $H = 2\,000$ : four observed, six censored. The exercise as prescribed asks for the four observed values to be perfectly ordered. Carried out literally, with the censored runs tied on both sides, it yields

Figure 7.  $\tau_{\text{ARF}}$  on the 100 seeds of each amplitude; the four medians of Table VI are annotated with their interval. The axis is symmetric logarithmic, linear below 1: a plain logarithmic axis would silently drop the 34 runs in which a tree is replaced at the change point itself ( $\tau_{\text{ARF}} = 0$ ), and the box would rest on 97 to 99 seeds while the title claims 100. Their count per amplitude is written in red on the zero line; they occur at all 20 amplitudes, three at the lowest and one at the highest. What to see: the median rises from the first amplitude to the second before it falls, the spread shrinks by more than an order of magnitude across the grid, and the two shaded amplitudes lie outside the decision domain.



$\rho = \tau_b = 1.000000$  exactly. The gap is zero and the counter-example demonstrates nothing.

The reason is sharper than “the example is badly chosen”. Ties on both sides require  $\tau_{\text{ARF}}$  itself to be censored, and on this campaign it never is: `censored_arf` = 0.0000 on all 2000 runs. The configuration under which the prescribed exercise produces no gap cannot occur in the data it is meant to illustrate.

Two things separate the three rows, and they should not be conflated. What removes the gap entirely is the tie structure: as soon as  $\tau_{\text{ARF}}$  is left uncensored, which is the situation that actually holds, the two coefficients part company by 0.071 even with the observed runs perfectly ordered. What then amplifies the gap is the ordering of the observed runs: with four anti-ordered observations, Spearman reports 0.751 where  $\tau_b$  reports 0.490, a gap of 0.261, nearly four times the ordered case.

The mechanism is the same in both rows. Spearman overstates the association because the six censored runs share a mid-rank that enters its sums of squares as an ordinary value, manufacturing agreement out of a tie block, while  $\tau_b$  removes those pairs from both counts. The prescribed exercise hides this by silently assuming ties on both sides, which cancels the effect exactly.

a) *Conclusion:* Pearson is excluded by definition rather than by degree, since censored runs carry no numerical value. Spearman is computable but overstates the association in the presence of tie blocks, and the tie blocks here are created by the censoring itself. Kendall’s  $\tau_b$  is used throughout what follows, with censored runs imputed to the last rank, which is legitimate because the horizon is common to all runs and only for that reason.

## VI. WHAT THE REPAIR DATE TELLS THE DETECTOR

Every coefficient here is Kendall’s  $\tau_b$  with censored runs imputed to the last rank, justified in Section V. Every confidence interval on a  $\tau_b$  is a bootstrap interval over the seeds of the stratum, 2000 resamples, quantiles 2.5% and 97.5%; every interval on a fraction is a Wilson 95% interval. Values are produced by `scripts/chiffres_QD.py` and `derives_QD.py`.

### A. Why Stratify

$\Delta e$  drives both sides of every correlation of interest: it shortens  $\tau_{\text{ARF}}$  and it raises the evidence available to the detector. A coefficient computed on the pooled 2000 runs therefore measures the common dependence on  $\Delta e$  rather than any association between the two quantities at a fixed operating point. Stratification removes that common cause. Each stratum holds 100 seeds, and the amplitude is constant inside it by construction.

### B. The Decision Rule, and Its Status

a) *A reservation, stated rather than glossed:* The protocol asks for a decision criterion written down before any correlation is inspected. That claim cannot honestly be made here. The pre-registered document of this work fixes the criteria of a *benchmark*: the tests that qualify candidate replacements for  $\tau_{\text{ARF}}$  against an analytic reference. It contains neither the indicator evaluated here, nor its comparators, nor the decision domain  $\Delta e \geq 0.10$ , nor the threshold on  $\tau_b$ , nor the rule on confidence-interval bounds. The columns `dans_domaine_decision`, `seuil_detectabilite` and `discernable` are produced by the same analysis pass as the coefficients they qualify. Anteriority is therefore not claimed for this criterion. It is fixed here, applied uniformly, and no cell is reclassified afterwards.

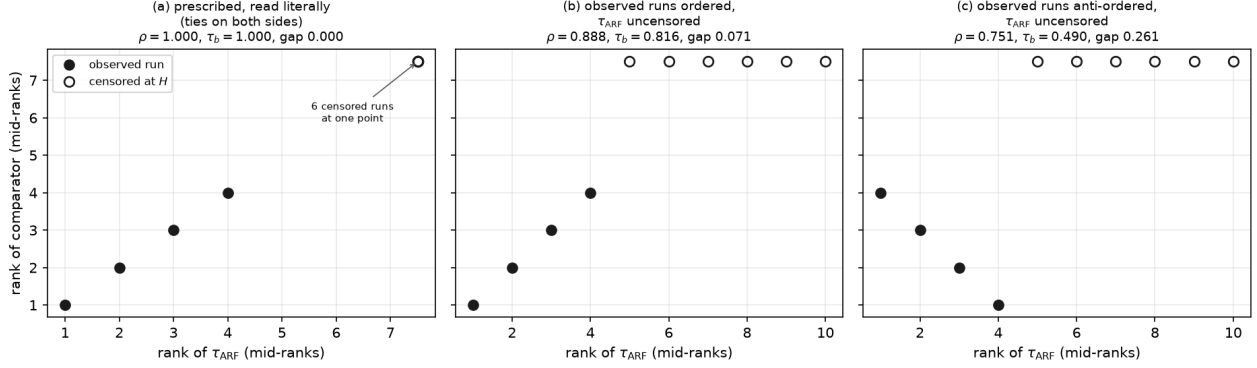
Table VII

TEN RUNS, FOUR OBSERVED AND SIX CENSORED AT  $H$ . EACH COEFFICIENT IS COMPUTED FROM ITS DEFINITION AND BY `SCIPY`; THE TWO AGREE TO  $2.2 \times 10^{-16}$  OR BETTER IN EVERY CELL, WHICH IS THE CONTROL ON THE CALCULATION.

| Configuration                                           | $\rho$ (Spearman) | $\tau_b$ (Kendall) | gap $\rho - \tau_b$ |
|---------------------------------------------------------|-------------------|--------------------|---------------------|
| Prescribed, read literally (ties both sides)            | 1.000000          | 1.000000           | 0.000000            |
| Observed values ordered, $\tau_{\text{ARF}}$ uncensored | 0.887625          | 0.816497           | 0.071129            |
| Observed values anti-ordered                            | 0.751068          | 0.489898           | 0.261170            |

Figure 8. The three configurations of Table VII, as mid-ranks. Filled markers: observed runs; hollow markers: runs censored at  $H$ , which share one mid-rank on the comparator side. (a) The prescribed exercise read literally also ties the six censored runs on the  $\tau_{\text{ARF}}$  side, and the two coefficients coincide. (b) With  $\tau_{\text{ARF}}$  left uncensored, the same four ordered observations already separate Spearman from Kendall by 0.071. (c) Anti-ordering the four observations widens the gap to 0.261. Table `QCD_contre_exemple_kendall`.

Ten fictitious runs, six censored: what separates Spearman from Kendall



*b) Two incompatible rules:* Two rules for calling an association discernible coexist in this work.

**Threshold rule.**  $> 0.13295637$ , the value in the `seuil_detectabilite` column, constant across the table. It is quoted exactly rather than rounded, since a rounded threshold would not reproduce the column.

**Interval rule.** The bootstrap confidence interval excludes zero. A  $\tau_b$  whose interval contains zero does not count.

They disagree on exactly three cells out of the 108 in the decision domain (Table VIII).

*c) The interval rule governs:* Three reasons, of unequal weight. A fixed threshold on  $|\tau_b|$  ignores the precision of the estimate, treating a coefficient of 0.13 measured on 100 seeds with a wide interval exactly like one measured tightly, whereas the interval rule uses what the bootstrap actually provides. The threshold is also a function of  $n$  alone, so it says nothing about the tie structure of the particular pair being tested, and Section V established that this structure differs systematically between the  $\tau_{\text{swap}}(q)$  comparators and the never-censored  $A(H)$  and  $S_{\text{max}}(H)$ . Finally, the pre-registered benchmark already uses the interval rule, so adopting it keeps one rule across the project instead of two.

### C. The Aggregation Collapse

For  $A(H)$  the pooled coefficient is  $+0.3965$  against a median stratified coefficient of  $+0.0526$ , a factor of 7.5. This is not a strong association weakened by stratification: the stratified coefficient is indistinguishable from zero under

this protocol at every one of the 18 amplitudes. The pooled coefficient manufactures a link rather than confirming a weak one. Mechanically it is Simpson's paradox on  $\Delta\epsilon$ : within each stratum there is nothing, and across strata both quantities move with the amplitude, so pooling recovers the amplitude and reports it as an association.

The effect is strongest exactly where the within-stratum association is weakest: ratio 7.54 for  $A(H)$  and 4.72 for  $\tau_{\text{swap}}(75\%)$ , but only 1.48 for  $S_{\text{max}}(H)$ , which does carry a genuine within-stratum signal. That is the diagnostic one should expect, and it is why no pooled coefficient is quoted as a result in this work.

### D. A Tautological Row, Marked as Such

At  $M = 10$  a quota of 10% requires exactly one tree, so  $\tau_{\text{swap}}(10\%) = \tau_{\text{ARF}}$  by (1). Its  $\tau_b$  against  $\tau_{\text{ARF}}$  is 1.000000 at all 20 amplitudes with confidence interval  $[1.000; 1.000]$ , by identity rather than by performance. The row is kept in Table IX and hatched in Figure 9 rather than deleted, so the reader can check that the instrumentation reproduces an identity it should reproduce. It carries no evidence about anything else.

A related caution applies to the other quotas. At  $M = 10$  neither  $q = 25\%$  nor  $q = 75\%$  is exactly attainable: 25% requires 3 trees, that is 30%, and 75% requires 8, that is 80%. The labels are kept for continuity with Section III-A, which uses the same convention, but they name the attainable quota above the nominal one.

Table VIII

THE THREE CELLS ON WHICH THE TWO RULES DISAGREE. NONE OF THE THREE IS QUOTED INDIVIDUALLY AS A RESULT ANYWHERE IN THIS REPORT.

| $\Delta e$ | comparator                 | $\tau_b$ | CI                   | threshold rule | interval rule |
|------------|----------------------------|----------|----------------------|----------------|---------------|
| 0.287      | $\tau_{\text{swap}}(75\%)$ | +0.1363  | $[-0.0068; +0.2704]$ | discernible    | not           |
| 0.287      | $S_{\text{max}}(H)$        | +0.1308  | $[+0.0027; +0.2519]$ | not            | discernible   |
| 0.391      | $\tau_{\text{swap}}(75\%)$ | +0.1240  | $[+0.0010; +0.2505]$ | not            | discernible   |

Figure 9.  $\tau_b(\tau_{\text{ARF}}, \cdot)$  for every comparator at every amplitude. All 120 coefficients are published, with no post hoc selection. The hatched row is tautological.

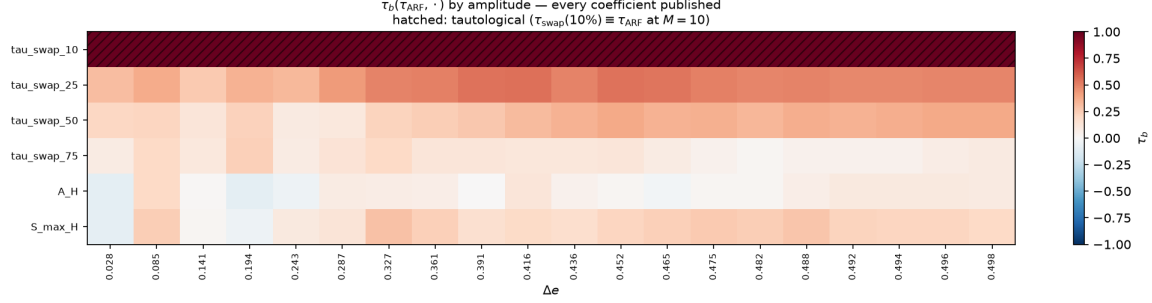


Table IX

POOLED AGAINST STRATIFIED COEFFICIENTS; THE STRATIFIED COLUMN IS THE MEDIAN OVER THE 18 AMPLITUDES OF THE DECISION DOMAIN. THE FIRST ROW IS TAUTOLOGICAL, SEE SECTION VI-D.

| Comparator                 | pooled $\tau_b$ | median stratified | ratio |
|----------------------------|-----------------|-------------------|-------|
| $\tau_{\text{swap}}(10\%)$ | +1.000000       | +1.000000         | 1.00  |
| $\tau_{\text{swap}}(25\%)$ | +0.644626       | +0.493224         | 1.31  |
| $\tau_{\text{swap}}(50\%)$ | +0.556494       | +0.336523         | 1.65  |
| $\tau_{\text{swap}}(75\%)$ | +0.461827       | +0.097934         | 4.72  |
| $A(H)$                     | +0.396512       | +0.052559         | 7.54  |
| $S_{\text{max}}(H)$        | +0.314673       | +0.212865         | 1.48  |

the evidence offered to the detector is attached to the wrong object if it rests on  $A(H)$  alone. The answer, stated sharply:

$\tau_{\text{ARF}}$  carries **no information** about the total accumulated excess error  $A(H)$ , indistinguishable from zero at all 18 amplitudes. It carries a **weak but real** association with  $S_{\text{max}}(H)$ , the quantity that determines whether the alarm fires: median  $\tau_b = +0.2129$ , discernible at 15 of 18 amplitudes, that is every amplitude from  $\Delta e = 0.287$  upwards without exception. Below  $\Delta e = 0.287$  the association is not distinguishable from zero either. Both figures are displaced towards zero by the tie correction (Section V), so the first is not an artefact of ties inflating a null and the second is if anything understated.

### E. Coefficients by Amplitude

### F. The Answer

The two quantities in Table X play different roles, and conflating them changes the answer.

$A(H)$  is the accumulated excess error over the horizon. Its association with  $\tau_{\text{ARF}}$  is indistinguishable from zero under this protocol at all 18 amplitudes: median +0.0526, every confidence interval straddling zero.

$S_{\text{max}}(H)$  is the peak of the detector path (2), in the closed form of Proposition 1. Its association is discernible at 15 of the 18 amplitudes, with a median of +0.2129, and the confidence interval lies entirely above zero at every amplitude from  $\Delta e = 0.287$  upwards. That count is the one the interval rule of Section VI-B gives; the discernible column implements the threshold rule and would give 14, the difference being the single cell  $\Delta e = 0.287$ , whose interval  $[+0.0027; +0.2519]$  excludes zero while its  $\tau_b$  of 0.1308 falls just under the threshold.

The distinction matters because it is  $S_{\text{max}}(H)$  that decides the alarm, not  $A(H)$ . The alarm is by definition the crossing of  $\lambda$  by  $S_{\text{max}}$ . A claim that  $\tau_{\text{ARF}}$  carries no information about

### G. Censoring: by Amplitude, and Whether It Drives Anything

Aggregate censoring rates conceal what matters, so they are reported by amplitude.

a)  $\tau_{\text{swap}}(75\%)$  *fails the group's own criterion.*: The criterion adopted for the benchmark requires censoring of at most 5% over the declared domain. Inside the decision domain,  $\tau_{\text{swap}}(75\%)$  reaches 54% at  $\Delta e = 0.141$ , exceeding the criterion by a factor of 10.8. Its worst value on the full grid is 78%, at  $\Delta e = 0.085$ , but that amplitude lies outside the decision domain and the figure is quoted as a grid measurement rather than as a breach of an in-domain criterion. Its coefficients are published above because suppressing them would hide the comparator that most stresses the method, but they are published with this failure stated: at  $\Delta e = 0.141$ , the lowest amplitude of the decision domain, the  $\tau_b$  of  $\tau_{\text{swap}}(75\%)$  rests on a minority of observed runs and is not quoted as evidence.

b) *No median delay beyond 50% censoring*: At  $\lambda = 25$  and  $\lambda = 50$  no median detection delay is reported anywhere in this report, and at  $\lambda = 8$  none is reported at  $\Delta e = 0.028$  where censoring reaches 90%. A median computed on the surviving

Figure 10. Forest plot of  $\tau_b(\tau_{\text{ARF}}, \cdot)$  with its bootstrap 95% confidence interval, for  $A(H)$  (left) and  $S_{\text{max}}(H)$  (right), one row per amplitude; filled squares mark intervals that exclude zero, hollow circles those that straddle it, and the two shaded rows lie outside the decision domain. 15 of 18 intervals exclude zero for  $S_{\text{max}}(H)$ , none for  $A(H)$ , under the interval rule of Section VI-B. The 15 are the 15 consecutive amplitudes from  $\Delta e = 0.287$  upward. Table QCD\_correlations\_stratifiedes\_full.

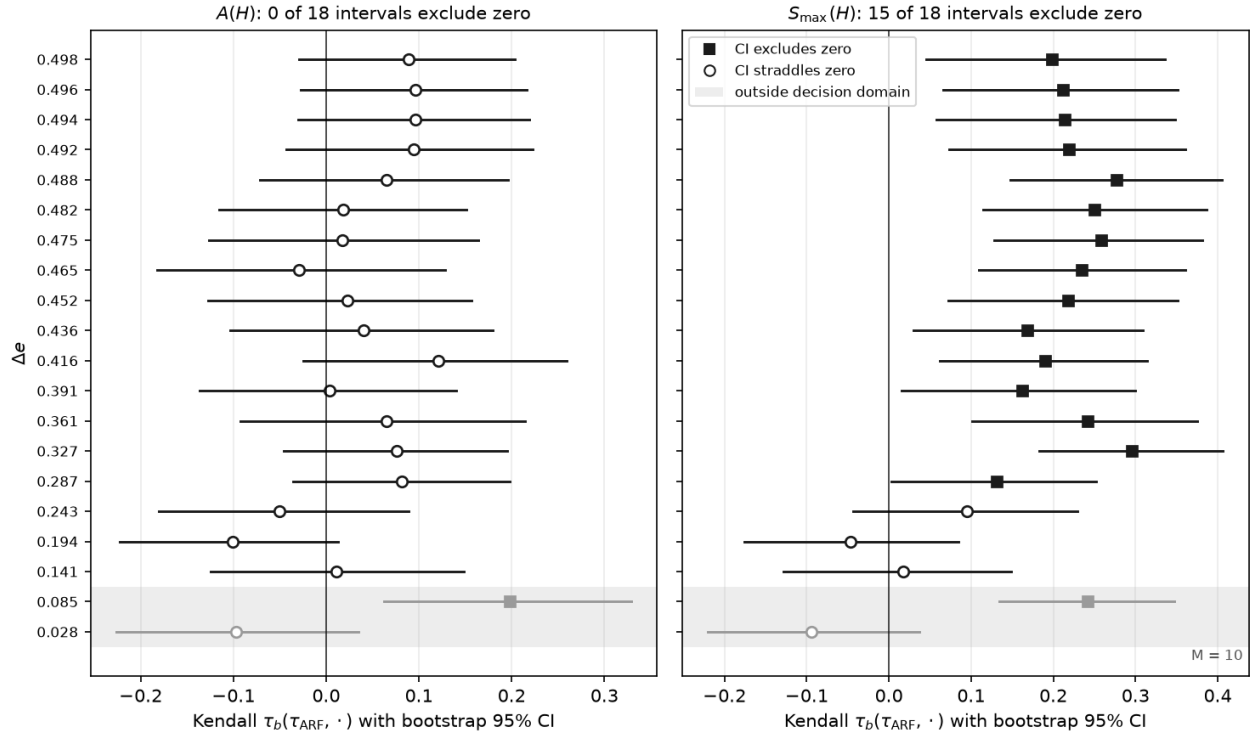
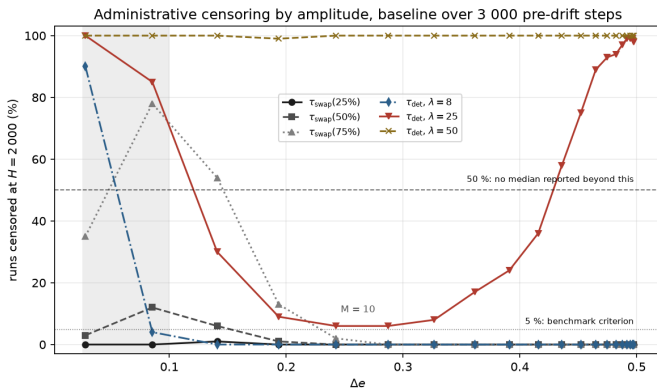


Figure 11. Censoring at  $H = 2000$  of every censorable duration, by amplitude, one series per duration, with the 50% line above which no median is reported and the 5% line of the benchmark criterion. What to see:  $\tau_{\text{swap}}(75\%)$  peaks at 78% at  $\Delta e = 0.085$  and is still above 50% at  $\Delta e = 0.141$ , the lowest amplitude of the decision domain, but only 35% at the very bottom of the grid, and it falls under 5% from  $\Delta e = 0.243$ ;  $\tau_{\text{det}}$  at  $\lambda = 25$  is U-shaped, below 10% only between  $\Delta e = 0.19$  and  $0.33$ ; at  $\lambda = 50$  it never leaves 99%. Tables QCD\_correlations\_stratifiedes\_full and QCD\_socle\_deux\_fenêtres.



runs excludes the slowest ones by construction and is not a median of the quantity of interest.

c) *Does censoring drive the results? No:* Each coefficient was recomputed twice, once with censored runs imputed to the last rank and once on complete cases only, then compared cell

by cell (Table XII, Figure 12). For the three never-censored comparators the two computations agree exactly, to 0.000000, which is what they must do and serves as a control on the implementation. For the two materially censored comparators the largest disagreement anywhere in the table is 0.0403, on  $\tau_{\text{swap}}(75\%)$ , an order of magnitude below the coefficients that carry the conclusions above and well inside their confidence intervals. The imputation convention of Section V-B could be replaced by complete-case analysis without changing a verdict.

d) *Why the campaign fixes no  $\lambda$ :* Table XIII gives the alarm behaviour of the three thresholds. At  $\lambda = 50$  a single run out of 2000 ever alarms, so with 99.95% censoring no correlation involving  $\tau_{\text{det}}$  is computable there at all. Had a threshold been frozen inside the simulation loop, as an earlier prototype did, the resulting table would carry a  $\tau_{\text{det}}$  column empty in all but one row. The transition from “the detector is never late” to “the detector never fires” happens entirely within a narrow band of the tuning parameter, which is what Section VII resolves.

## H. Reservations, and What Is Not Addressed

a) *Multiplicity:* The decision domain contains 108 cells, of which 68 are significant at the 5% level with no correction for multiplicity. The signal for  $S_{\text{max}}(H)$  above  $\Delta e = 0.287$  is far too systematic to be a multiplicity artefact, running to 15 consecutive amplitudes in the same direction (Figure 10), but

Table X  
 $\tau_b(\tau_{\text{ARF}}, \cdot)$  WITH BOOTSTRAP CONFIDENCE INTERVALS OVER THE SEEDS, ON THE 18 AMPLITUDES OF THE DECISION DOMAIN.

| $\Delta e$ | $A(H)$   |         |         | $S_{\max}(H)$ |         |         |
|------------|----------|---------|---------|---------------|---------|---------|
|            | $\tau_b$ | CI low  | CI high | $\tau_b$      | CI low  | CI high |
| 0.141      | +0.0111  | -0.1249 | +0.1485 | +0.0174       | -0.1288 | +0.1487 |
| 0.194      | -0.1009  | -0.2228 | +0.0127 | -0.0470       | -0.1759 | +0.0846 |
| 0.243      | -0.0504  | -0.1802 | +0.0887 | +0.0952       | -0.0432 | +0.2294 |
| 0.287      | +0.0815  | -0.0357 | +0.1978 | +0.1308       | +0.0027 | +0.2519 |
| 0.327      | +0.0762  | -0.0460 | +0.1954 | +0.2954       | +0.1832 | +0.4063 |
| 0.361      | +0.0651  | -0.0924 | +0.2146 | +0.2413       | +0.1009 | +0.3748 |
| 0.391      | +0.0037  | -0.1366 | +0.1398 | +0.1623       | +0.0154 | +0.3002 |
| 0.416      | +0.1212  | -0.0245 | +0.2595 | +0.1904       | +0.0621 | +0.3141 |
| 0.436      | +0.0402  | -0.1037 | +0.1796 | +0.1682       | +0.0295 | +0.3088 |
| 0.452      | +0.0230  | -0.1277 | +0.1570 | +0.2177       | +0.0722 | +0.3511 |
| 0.465      | -0.0296  | -0.1820 | +0.1279 | +0.2347       | +0.1095 | +0.3611 |
| 0.475      | +0.0171  | -0.1260 | +0.1642 | +0.2584       | +0.1282 | +0.3817 |
| 0.482      | +0.0179  | -0.1151 | +0.1509 | +0.2496       | +0.1141 | +0.3861 |
| 0.488      | +0.0650  | -0.0710 | +0.1963 | +0.2764       | +0.1478 | +0.4055 |
| 0.492      | +0.0945  | -0.0432 | +0.2226 | +0.2193       | +0.0736 | +0.3608 |
| 0.494      | +0.0964  | -0.0301 | +0.2188 | +0.2135       | +0.0577 | +0.3478 |
| 0.496      | +0.0966  | -0.0273 | +0.2164 | +0.2122       | +0.0659 | +0.3518 |
| 0.498      | +0.0890  | -0.0293 | +0.2038 | +0.1985       | +0.0457 | +0.3359 |

Table XI

CENSORING OVERALL, AT ITS WORST AMPLITUDE ON THE FULL GRID, AND AT ITS WORST AMPLITUDE INSIDE THE DECISION DOMAIN. THE COLUMNS DIFFER BECAUSE THE TWO AMPLITUDES BELOW  $\Delta e = 0.10$  CARRY MOST OF THE CENSORING AND LIE OUTSIDE THE DOMAIN ON WHICH THE CRITERION IS DECLARED. BASELINE ESTIMATED OVER 3 000 PRE-DRIFT STEPS (SECTION IV-C). AT  $\lambda = 50$  THE SINGLE RUN THAT ALARMS IS AT  $\Delta e = 0.194$ .

| Quantity                          | overall | worst on grid (at $\Delta e$ )      | worst in domain (at $\Delta e$ )    |
|-----------------------------------|---------|-------------------------------------|-------------------------------------|
| $\tau_{\text{ARF}}$               | 0.0000  | 0.0000                              | 0.0000                              |
| $A(H), S_{\max}(H)$               | 0.0000  | 0.0000                              | 0.0000                              |
| $\tau_{\text{swap}}(25\%)$        | 0.0005  | 0.0100 (0.141)                      | 0.0100 (0.141)                      |
| $\tau_{\text{swap}}(50\%)$        | 0.0110  | 0.1200 (0.085)                      | 0.0600 (0.141)                      |
| $\tau_{\text{swap}}(75\%)$        | 0.0910  | 0.7800 (0.085)                      | 0.5400 (0.141)                      |
| $\tau_{\text{det}}, \lambda = 8$  | 0.0470  | 0.9000 (0.028)                      | 0.0000 (all 18)                     |
| $\tau_{\text{det}}, \lambda = 25$ | 0.6110  | 1.0000 (0.028)                      | 0.9900 (0.492, 0.494, 0.496)        |
| $\tau_{\text{det}}, \lambda = 50$ | 0.9995  | 1.0000 (every $\Delta e$ but 0.194) | 1.0000 (every $\Delta e$ but 0.194) |

Table XII

SENSITIVITY OF  $\tau_b$  TO THE TREATMENT OF CENSORED RUNS. AN EARLIER INTERNAL NOTE CLAIMED THE TWO COMPUTATIONS AGREE TO THREE DECIMAL PLACES; THAT IS TOO STRONG. IT HOLDS FOR  $\tau_{\text{swap}}(25\%)$  AND FOR THE NEVER-CENSORED COMPARATORS, WHILE THE LAST TWO ROWS DIFFER AT THE SECOND DECIMAL PLACE, AND IT IS THESE TWO NUMBERS THAT ARE QUOTED.

| Comparator                                    | max  imputed – complete cases |
|-----------------------------------------------|-------------------------------|
| $\tau_{\text{swap}}(10\%), A(H), S_{\max}(H)$ | 0.000000                      |
| $\tau_{\text{swap}}(25\%)$                    | 0.000228                      |
| $\tau_{\text{swap}}(50\%)$                    | 0.030591                      |
| $\tau_{\text{swap}}(75\%)$                    | 0.040259                      |

individual cells whose  $|\tau_b|$  is near the detectability threshold are not quoted on their own.

*b) The prescribed return time is not correlated run by run:*

The official statement of the question asks for the correlation of  $\tau_{\text{ARF}}$  with the residual error and with the effective time of return to normal. The second is not computed as a per-run correlation anywhere, and the reason belongs here rather than in silence.  $\tau_{\text{rec}}$  is defined only on the seed-averaged error curve,

Table XIII

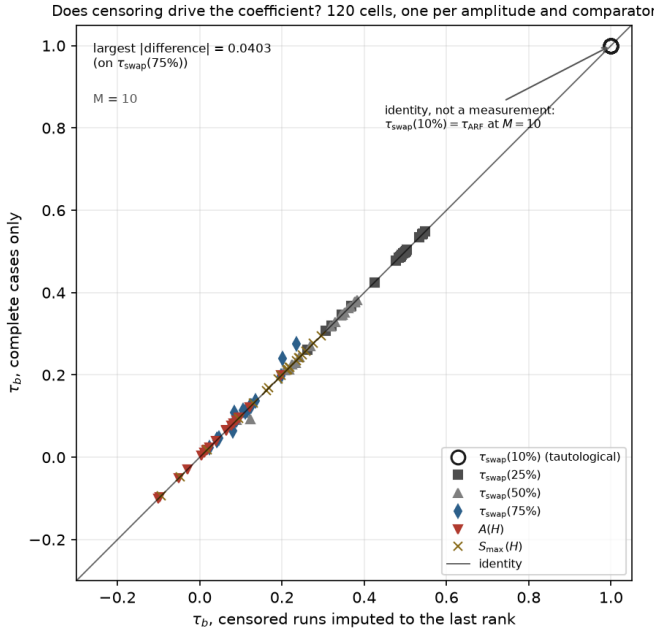
ALARM BEHAVIOUR BY THRESHOLD. WHO WINS THE RACE IN THOSE RUNS IS TABLE V AND, POOLED OVER THE GRID, TABLE II.

| $\lambda$ | runs that alarm | silent |
|-----------|-----------------|--------|
| 8         | 1 906/2 000     | 4.70%  |
| 25        | 778/2 000       | 61.10% |
| 50        | 1/2 000         | 99.95% |

one value per amplitude: on a single run the binary sequence is far too noisy for a threshold crossing to signal genuine recovery. The correlations above are computed run by run, so the two objects are not commensurable. It appears in Figure 3 as the  $\tau_{\text{rec}}$  marker, the only use its definition supports, dashed at the amplitudes where the test has no power to resolve it.

“Return of the error to an acceptable level” also fails as an adaptation criterion, for the reason Section III-C establishes. At  $\Delta e = 0.498$  the threshold  $\hat{p}_0 + \rho \Delta e$  is crossed at step 46, with or without the persistence requirement, because the post-drift problem is easier than the original one and the error

Figure 12. The 120 cells of Table XII one by one:  $\tau_b$  on complete cases against  $\tau_b$  with censored runs imputed to the last rank, one marker per comparator, with the identity line. The hollow circle at (1, 1) is  $\tau_{\text{swap}}(10\%)$ , an identity at  $M = 10$  and not a measurement. Table QCD\_cas\_complets\_full.



falls below the pre-drift baseline. A criterion that a simplified problem satisfies immediately does not measure adaptation.

c) *Scope*: Everything above is measured at  $M = 10$  on one campaign, so no result of this question holds as a function of forest size; the instrumented runs at other values of  $M$  were produced for a different test and carry no evidence trace. No result involves  $\tau_{\text{det}}$  at  $\lambda = 25$  or  $\lambda = 50$  as a correlated variable, for the censoring reason given above. Harrell’s  $C$  index was not used, because the common-horizon argument of Section V-B makes rank imputation licit here and the complete-case comparison shows the choice is immaterial; under a variable horizon that reasoning would fail and the  $C$  index would be required.

## VII. THRESHOLD CHOICE AND THE NO-DRIFT CONTROL

This section collects two results that belong to no single question of the subject and that would distort the one they were attached to. First, detecting a drift and winning the race against the forest are different things, and they separate as soon as the threshold rises — which is what the narrow transition band of Section VI-G looks like once it is resolved. Second, the forest replaces trees at a steady rate when nothing changes at all, which disqualifies the date of the first replacement as a signal on its own. Everything below is recomputed offline from the recorded error trajectories by `resultats_R2/scripts/chiffres_QE.py`; no campaign is rerun.

a) *Control*:  $S_t$  does not depend on  $\lambda$ , so a single offline pass over each trajectory yields the first crossing for every threshold. The sweep must reproduce the campaign exactly at the three thresholds of the subject, and it does: the alarm

counts of Table XIII are recovered to the run. Nothing below is readable without this check.

### A. Detecting Is Not Winning

The campaign of 2000 runs is read at six thresholds. The drift-free arm — 100 runs with no change point at all, same horizon — gives what the same threshold costs when nothing happens.

a) *The two columns come apart*: At  $\lambda = 15$  the alarm still fires in 92.9% of runs, barely below the 95.3% of  $\lambda = 8$ , but it arrives first in only 36.3% of them. Almost two thirds of its alarms are raised after the forest has already replaced a tree — the *Zombie alarms* of the manuscript. Reading the detection column alone would suggest that a moderate threshold escapes the blind spot. It does not.

b) *The paradoxical signature survives at  $\lambda = 15$* : Amplitude by amplitude, the share of runs won by the detector falls as the drift grows, which is the signature the manuscript reports at  $\lambda = 25$ .

At  $\lambda = 8$  the rate climbs with the amplitude, from 62% to 99%. At  $\lambda = 15$  it peaks at 81% then falls to 18%, and at  $\lambda = 20$  it collapses to 1%. The transition  $\lambda^* \approx 15$  to 25 announced in the manuscript is therefore confirmed on the race, not contradicted.

c) *What the false-alarm column adds*: The manuscript grants that a hyper-reactive threshold reaches a safe zone but states that it “sacrifices false-alarm robustness”. On this bench that cost is measured rather than assumed:  $\lambda = 8$  raises an alarm in 5 of 100 runs where nothing changes, which is the calibration level usually adopted for drift detectors. So on the Bernoulli bench the only threshold that wins the race does so at an ordinary false-alarm cost, and the admissible set is not empty here.

d) *Three reservations*: The false-alarm rate is measured over 2000 steps on 100 runs; over a longer deployment such alarms accumulate, and the upper end of the Wilson interval at  $\lambda = 8$  is 11.2%. The bench is homoscedastic, so this says nothing about the *Fundamental Tension* the manuscript establishes on ARMA–GARCH streams, where bounding false alarms under clustered volatility is what empties the admissible set. And at the weakest amplitude no threshold wins a single run, which Section I establishes without any dependence assumption.

### B. The No-Drift Control

The same drift-free arm answers a question that needs no threshold at all: what does  $\tau_{\text{ARF}}$  read when there is nothing to repair? Sections III-A and IV both read this arm in passing; it is reported here in full.

a) *Reading*: Replacement churn under pure noise is the normal regime of the ARF, not an anomaly: nine trees out of ten are renewed over two thousand steps with no drift whatsoever. At the weakest amplitude of the grid the stopwatch reads 118 steps against 105 with no drift at all, so at that amplitude it does not distinguish a real change from its absence. This is the control any candidate replacement for  $\tau_{\text{ARF}}$  must pass before

Table XIV

SIX THRESHOLDS ON THE SAME CAMPAIGN. “WINS THE RACE” IS  $\tau_{\text{det}} < \tau_{\text{ARF}}$  OVER ALL 2 000 RUNS, CENSORED RUNS COUNTED AS LOSSES; THE NEXT COLUMN RESTRICTS TO THE RUNS THAT ALARM. BRACKETS: WILSON 95% INTERVAL,  $n = 100$  DRIFT-FREE RUNS. THE ROWS AT  $\lambda = 8, 25$  AND 50 REPRODUCE TABLES V AND XIII.

| $\lambda$ | alarm fires | wins the race | of uncensored | false alarms, no drift   |
|-----------|-------------|---------------|---------------|--------------------------|
| 8         | 95.3%       | <b>91.0%</b>  | 95.4%         | 5/100 = 5.0% [2.2; 11.2] |
| 12        | 94.3%       | 78.8%         | 83.5%         | 2/100 = 2.0% [0.6; 7.0]  |
| 15        | 92.9%       | <b>36.3%</b>  | 39.1%         | 0/100 = 0.0% [0.0; 3.7]  |
| 20        | 66.2%       | 8.1%          | 12.2%         | 0/100 = 0.0% [0.0; 3.7]  |
| 25        | 38.9%       | 2.5%          | 6.6%          | 0/100 = 0.0% [0.0; 3.7]  |
| 50        | 0.1%        | 0.0%          | 0.0%          | 0/100 = 0.0% [0.0; 3.7]  |

Table XV

SHARE OF RUNS WON BY THE DETECTOR, SIX AMPLITUDES SPANNING THE GRID, OVER ALL 100 RUNS OF THE CELL. THE  $\lambda = 8$  COLUMN IS THE LAST COLUMN OF TABLE V, RESTRICTED TO THESE SIX ROWS.

| $\Delta e$ | $\lambda = 8$ | $\lambda = 15$ | $\lambda = 20$ |
|------------|---------------|----------------|----------------|
| 0.028      | 0%            | 0%             | 0%             |
| 0.085      | 62%           | 38%            | 20%            |
| 0.141      | 88%           | 81%            | 49%            |
| 0.243      | 97%           | 71%            | 21%            |
| 0.361      | 98%           | 44%            | 3%             |
| 0.498      | 99%           | 18%            | 1%             |

Table XVI

THE DRIFT-FREE ARM AGAINST THE CAMPAIGN; QUANTILES OF THE FIRST REPLACEMENT, 43 AND 172 STEPS, AND TREES COUNTED OVER THE HORIZON  $H$ . THE MEDIAN AT  $\Delta e = 0.028$  CARRIES THE BOOTSTRAP INTERVAL [92.5; 158.5] OF TABLE VI.

| On 100 runs with no change point                              |           |
|---------------------------------------------------------------|-----------|
| runs where at least one tree is replaced                      | 100/100   |
| first replacement, median                                     | 105 steps |
| distinct trees renewed, median                                | 9/10      |
| runs whose forest is entirely renewed                         | 24/100    |
| median $\tau_{\text{ARF}}$ at $\Delta e = 0.028$ (real drift) | 118 steps |
| median $\tau_{\text{ARF}}$ over the whole grid                | 38 steps  |

its correlations are worth computing, and it is test A1 of the benchmark.

### C. What This Section Establishes

Detection and victory separate: a threshold can alarm on almost every drift and still arrive last on two thirds of them, so a detection rate quoted alone does not describe the blind spot. On this bench one threshold does win the race,  $\lambda = 8$ , at a false-alarm cost of 5% per two thousand drift-free steps, which restricts the manuscript’s “no CUSUM threshold escapes it” to the settings it actually tested here. And the date of the first replacement fires as readily without drift as with the weakest one, which is a property of the indicator rather than of the detector — the same verdict Section VI reaches from the association side.